DISTRICT PANCHAYATH KASARAGOD

EQIP 2026

Educational Quality Improvement Programme for class -X

PHYSICS

ENGLISH MEDIUM

Standard -X

QUESTION POOL



DISTRICT INSTITUTE OF EDUCATION AND TRAINING (DIET)
KASARAGOD

EQIP 2025-26

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ആശംസ

വികേന്ദ്രീകൃത ആസൂത്രണത്തിലൂടെയും നിർവ്വഹണത്തിലൂടെയും കേരളത്തിലെ ആരോഗ്യവിദ്യാഭ്യാസ മേഖലകളെ ദേശീയ തലത്തിൽ ഒന്നാമതെത്തിക്കാൻ നമുക്ക് കഴിഞ്ഞിട്ടുണ്ട്. ഈ നേട്ടങ്ങൾ കൈവരിക്കാൻ പ്രാദേശിക ഭരണകൂടങ്ങൾ സ്തുത്യർ ഹമായ പങ്കുവഹിച്ചു. ദേശീയ സംസ്ഥാനതല പഠനങ്ങൾ നമ്മുടെ കുട്ടികളുടെ പഠന നിലവാരം ഇനിയും ഉയരേണ്ടതുണ്ട് എന്ന സൂചനയാണ് നൽകുന്നത്. പത്താം ക്ലാസിലെ കുട്ടികളുടെ പഠനപ്രശ്നങ്ങൾ പരിഹരിക്കാൻ കഴിഞ്ഞ കുറെ വർഷങ്ങളായി കാസറഗോഡ് ജില്ലാ പഞ്ചായത്തിന്റെ നേതൃത്വത്തിൽ പൊതുവിദ്യാഭ്യാസ വകുപ്പും കാസറഗോഡ് ഡയറ്റും ഒത്തു ചേർന്ന് നടപ്പാക്കുന്ന **'എക്വിപ്പ്' 2025-26 (EQIP 2025-26)** പഠനപരിപോഷണ പരിപാടിക്ക് എല്ലാ പിന്തുണയും ഉറപ്പുതരുന്നു. പൊതുപരീക്ഷകളെ അഭിമുഖീകരിക്കുന്ന പത്താം ക്ലാസിലെ കുട്ടികളുടെ പഠനപ്രവർത്തനങ്ങളുടെ മികവിന്റെ അടയാളമായി മാറുകയാണ് വാർഷിക പരീക്ഷകൾ. അറിവിന്റെ തെളിമയോടെ ഓരോ വിദ്യാർത്ഥിക്കും പരീക്ഷ എഴുതാൻ കഴിയണം. വിദ്യാർത്ഥികളുടെ ജീവിതത്തിലെ ഏറ്റവും പ്രധാനപ്പെട്ട പരീക്ഷകൾക്ക് വേണ്ടി തയ്യാറാക്കിയ ചോദ്യബാങ്കിന് എല്ലാവിധ ആശംസകളും നേരുന്നു. നന്നായി പഠിക്കുക. പരീക്ഷയെ സധൈര്യം നേരിടുക. തളരാതെ മുന്നോട്ട്. വിജയം നിങ്ങളോടൊപ്പമുണ്ട്. ആശംസകൾ.

ശ്രീമതി ബേബി ബാലകൃഷ്ണൻ
ജില്ലാ പഞ്ചായത്ത് പ്രസിഡന്റ്
കാസറഗോഡ്



ആശംസ

കാസറഗോഡ് ജില്ലാ പഞ്ചായത്തിന്റെ നേതൃത്വത്തിൽ ജില്ലയിലെ പൊതുവിദ്യാഭ്യാസ മേഖലയെ ശക്തിപ്പെടുത്തുന്നതിന് നിരവധി പ്രവർത്തനങ്ങളാണ് നടന്നുവരുന്നത്. പൊതുവിദ്യാഭ്യാസ മേഖലയെ പൂർവ്വാധികം കരുത്തോടെ നാം മുന്നോട്ട് നയിക്കുകയാണ്. ഈ ഘട്ടത്തിലാണ് കാസറഗോഡ് ജില്ലാ പഞ്ചായത്തും, പൊതുവിദ്യാഭ്യാസ വകുപ്പും, വിദ്യാഭ്യാസ പരിശീലന കേന്ദ്രവും **(DIET)** പത്താം ക്ലാസ് വിദ്യാർത്ഥികളുടെ പഠന വിടവുകൾ പരിഹരിക്കുന്നതിനും ആത്മവിശ്വാസത്തോടെ പൊതുപരീക്ഷയെ നേരിടാൻ അവരെ പ്രാപ്തരാക്കുന്നതിനും വേണ്ടി പഠനപരിപോഷണ സാമഗ്രി തയ്യാറാക്കിയിട്ടുള്ളത്. ഈ പദ്ധതിയുടെ ഭാഗമായി ആത്യന്തിക വിലയിരുത്തലിനെ അഭിമുഖീകരിക്കാൻ കുട്ടികളെ സഹായിക്കുന്ന തരത്തിൽ വിവിധ വിഷയങ്ങളിൽ വിലയിരുത്തൽ സമീപനത്തിനനുസരിച്ചുള്ള ചോദ്യശേഖരമാണ് **(Question Pool)** ഈ അക്കാദമിക വർഷം തയ്യാറാക്കിയിട്ടുള്ളത്. കുട്ടികളുടെ അക്കാദമിക് മികവ് ഉറപ്പുവരുത്തിക്കൊണ്ട് മികച്ച ഗ്രേഡുകൾ നേടാൻ അവരെ സജ്ജമാക്കാൻ **എക്വിപ് 2025-26 (EQIP 2025-26)** എന്ന പേരിൽ തയ്യാറാക്കിയ ഈ പദ്ധതിക്ക് കഴിയട്ടെയെന്ന് ആശംസിക്കുന്നു.

സ്നേഹപൂർവ്വം

അഡ്വ. സരിത എസ്.എൻ.

ആരോഗ്യ-വിദ്യാഭ്യാസ സ്ഥിരം സമിതി അധ്യക്ഷ,

ജില്ലാ പഞ്ചായത്ത്, കാസറഗോഡ്



ആശംസ

ജില്ലയിലെ അക്കാദമിക പ്രവർത്തനങ്ങളെ ഏകോപിപ്പിച്ച് മുന്നോട്ട് നയിക്കുന്നതിന്റെ പ്രധാന ഉത്തരവാദിത്തം കാസറഗോഡ് ജില്ലാ വിദ്യാഭ്യാസ പരിശീലന കേന്ദ്രത്തിന്റെ (DIET) നേതൃത്വത്തിൽ കാലങ്ങളായി ചെയ്തുവരുന്നുണ്ട്. പ്രീ-പ്രൈമറി തലം മുതൽ ഹയർസെക്കണ്ടറി തലംവരെയുള്ള മേഖലകളിൽ വിവിധങ്ങളായ പദ്ധതികൾ ആസൂത്രണം ചെയ്യാനും നിർവഹിക്കാനും ഡയറ്റിന് ഇതിനകം കഴിഞ്ഞിട്ടുണ്ട്. പത്താം ക്ലാസ്സിലെ കുട്ടികളുടെ പഠനപ്രശ്നങ്ങൾ പരിഹരിക്കാൻ കഴിഞ്ഞ കുറച്ചു വർഷങ്ങളായി ഡയറ്റിന്റെ നേതൃത്വത്തിൽ വിദ്യാഭ്യാസ വകുപ്പ് നടപ്പിലാക്കുന്ന പദ്ധതിയാണ് **എക്വിപ് (Educational Quality Improvement Programme).** അതാതു വർഷത്തെ കുട്ടികളുടെ പഠനപ്രശ്നങ്ങൾ പരിഗണിച്ചുകൊണ്ടാണ് പ്രവർത്തനങ്ങൾ ആസൂത്രണം ചെയ്യുന്നത്. പത്താംതരം പാഠപുസ്തകവും വിലയിരുത്തൽ രീതിയും മാറിയ ഘട്ടത്തിൽ ആണല്ലോ നമ്മൾ. അതുകൊണ്ട് ഈ പദ്ധതിയുടെ ഭാഗമായി പത്താംതരത്തിലെ ആത്യന്തിക വിലയിരുത്തലിനെ അഭിമുഖീകരിക്കാൻ കുട്ടികളെ സഹായിക്കുന്ന തരത്തിൽ വിവിധ വിഷയങ്ങളിൽ പുതിയ വിലയിരുത്തൽ സമീപനത്തിനനുസരിച്ച് ചോദ്യശേഖരം (Question Pool) പരിചയപ്പെടുത്താനാണ് ഇപ്രാവശ്യം ഉദ്ദേശിക്കുന്നത്. ജില്ലാ പഞ്ചായത്തിന്റെ നേതൃത്വത്തിൽ ഡയറ്റിന്റെ അക്കാദമിക സഹായത്തോടെ മലയാളത്തിലും ഇംഗ്ലീഷിലും കന്നഡയിലും പത്താം തരത്തിലെ ഐ.ടി. ഒഴികെയുള്ള വിഷയങ്ങളിൽ ചോദ്യശേഖരം തയ്യാറാക്കി നൽകാനാണ് ഉദ്ദേശിക്കുന്നത്. ഈ അധ്യയനവർഷം പത്താംതരം പരീക്ഷ എഴുതുന്ന മുഴുവൻ കുട്ടികൾക്കും ഈ പദ്ധതിയുടെ പ്രയോജനം ലഭിക്കുമെന്ന് പ്രതീക്ഷിക്കുന്നു. വ്യത്യസ്ത പഠനശേഷിയും പഠനമികവുമുള്ള എല്ലാ വിഭാഗം കുട്ടികൾക്കും ഈ സാമഗ്രി പ്രയോജനപ്പെടട്ടെ എന്ന് ആശംസിക്കുന്നു. ഒരു കൂട്ടം അധ്യാപകരുടെ ആത്മാർത്ഥമായ പ്രവർത്തനമാണ് ഈ സാമഗ്രി വികസിപ്പിക്കാൻ സഹായകമായത്. അധ്യാപകരുടെ ആത്മാർത്ഥമായ പിന്തുണയും പ്രോത്സാഹനവും അനിവാര്യമായ ഈ ഉദ്യമത്തിൽ എല്ലാവരുടെയും സഹായസഹകരണങ്ങൾ തുടർന്നും പ്രതീക്ഷിച്ചുകൊണ്ട് ഏവർക്കും വിജയാശംസകൾ നേരുന്നു. ആശംസകളോടെ,

ഡോ. രഘുരാമ ഭട്ട് കെ.
പ്രിൻസിപ്പാൾ
ഡയറ്റ് കാസറഗോഡ്



ആമുഖം

കാസറഗോഡ് ജില്ലാ പഞ്ചായത്തിന്റെ നേതൃത്വത്തിൽ ജില്ലയിലെ പൊതുവിദ്യാഭ്യാസം ശക്തിപ്പെടുത്തുന്നതിന് വ്യത്യസ്തങ്ങളായ നിരവധി പ്രവർത്തനങ്ങൾ നടന്നുവരികയാണ്. അതേസമയം ദേശീയ-സംസ്ഥാന പഠനങ്ങൾ നമ്മുടെ ജില്ലയിലെ കുട്ടികളുടെ പ്രകടനം ഇനിയും മെച്ചപ്പെടേണ്ടതുണ്ട് എന്ന സൂചനയാണ് നൽകുന്നത്. ഈ പശ്ചാത്തലത്തിലാണ് ജില്ലയിലെ പൊതുവിദ്യാലയങ്ങളിൽ നിന്ന് **2025-26** അധ്യയന വർഷം എസ്.എസ്.എൽ.സി. പരീക്ഷകൾ അഭിമുഖീകരിക്കുന്ന കുട്ടികൾക്ക് പഠനപിന്തുണ നൽകുന്നതിന് ജില്ലാ പഞ്ചായത്തിന്റെയും പൊതുവിദ്യാഭ്യാസ വകുപ്പിന്റെയും സംയുക്താഭിമുഖ്യത്തിൽ വ്യത്യസ്ത വിഷയങ്ങളിൽ പഠനസാമഗ്രികൾ തയ്യാറാക്കേണ്ടതിന്റെ ആവശ്യകത ജില്ലാതല ഉന്നതാധികാര യോഗങ്ങളിൽ ചർച്ചചെയ്യപ്പെട്ടത്. ഇതിന്റെ അടിസ്ഥാനത്തിൽ ഡയറ്റ് കാസറഗോഡിന്റെ അക്കാദമിക നേതൃത്വത്തിൽ ജില്ലയിലെ മികച്ച അധ്യാപകരെ ഉൾപ്പെടുത്തിക്കൊണ്ട് പത്താംതരത്തിൽ ഐ.ടി. ഒഴിച്ചുള്ള എല്ലാ വിഷയങ്ങളിൽ വിലയിരുത്തൽ സമീപനത്തിലനുസരിച്ച് ചോദ്യശേഖരം **(Question Pool)** തയ്യാറാക്കിയിരിക്കുകയാണ്. ജില്ലയിൽ നിന്നും പൊതുപരീക്ഷയെ അഭിമുഖീകരിക്കുന്ന മുഴുവൻ എസ്.എസ്.എൽ.സി, വിദ്യാർത്ഥികൾക്കും ആത്മവിശ്വാസം വളർത്തുന്നതിനും ഉന്നത വിജയം നേടുന്നതിനും ഈ ഉദ്യമം സഹായകമാകട്ടെയെന്ന് ആത്മാർത്ഥമായി ആഗ്രഹിക്കുന്നു. ഈ പദ്ധതിയെ നെഞ്ചേറ്റിയ പ്രിയപ്പെട്ട അധ്യാപക സുഹൃത്തുക്കൾക്ക് ഈ പുസ്തകത്തെ ഫലപ്രദമായി ഉപയോഗിക്കാൻ കഴിയട്ടെ.

എല്ലാവർക്കും വിജയാശംസകൾ.

ശ്രീ. മധുസൂദനൻ ടി.വി.
ജില്ലാ വിദ്യാഭ്യാസ ഉപഡയറക്ടർ
കാസറഗോഡ്

PHYSICS

English Medium

Chapter 1

SOUND WAVES

1. Select the factors that affect the natural frequency of an object are (1)

   i) Length of the object

   ii) Size of the object

   iii) Elasticity

   iv) Nature of the material

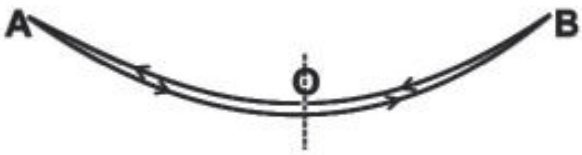
   a) Only i, ii, iii are correct

   b) Only ii, iii are correct

   c) Only i, ii, iv are correct

   d) All are correct

2. The path of a pendulum is depicted. Choose the situation in which one oscillation is completed from the following statements. (1)



   i) If it starts from A and returns to A

   ii) If it starts from O and reaches B and returns to A and then returns to O.

   iii) If it starts from O and reaches B and returns to O.

   iv) If it starts from B and returns to B

   a) Only i, iii, iv are correct

   b) Only i, ii, iv are correct

   c) Only i, ii are correct

   d) Only iii, iv are correct

3. Statement : When the length of the string of a simple pendulum is reduced, the number of oscillations increases.

Reason : When the length of the pendulum increases, frequency decreases (1)

Which of the following options is correct?

a) The statement and the reason are correct, the reason explains the statement.

b) The statement and the reason are correct, the reason does not explain the statement.

c) The statement is correct and the reason is wrong.

d) The statement is wrong and the reason is correct.

4. Pick out the correct option that is suitable for transverse waves.

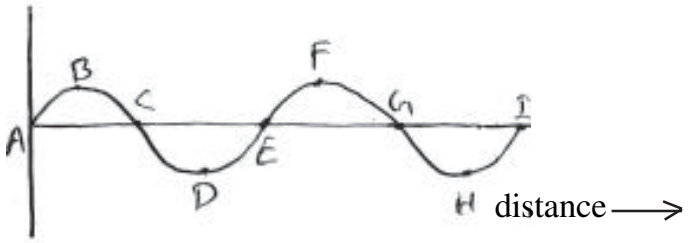
i) Microwave ii) Radio wave

iii) Light wave iv) Sound wave

a) i, ii, iii b) i, ii, iv

c) ii, iii d) i, ii

5. Graphic representation of a wave is given. Choose the one that does not indicate the wavelength? (1)



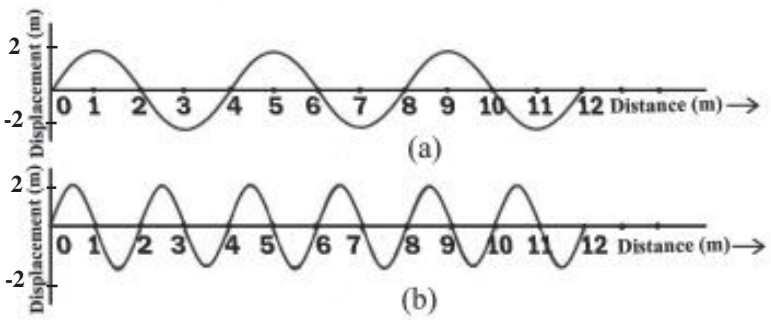
a) A to E

b) B to E

c) C to G

d) D to H

6.



The time taken by the waves shown in the figure to travel from A to B is 0.3 second. Which of the following is the same for both waves? (1)

a) Wave Speed, Wavelength
b) Frequency, Amplitude
c) Wave Speed, Amplitude
d) Wavelength, Amplitude

7. Statement : The sound of the Galton whistle cannot be heard by humans.

Reason : The Galton whistle produces infrasonic sound. (1)

Which of the following is correct?

a) The statement and the reason are correct. The reason explains the statement.
b) The statement and the reason are correct. The reason does not explain the statement
c) The statement is correct but the reason is incorrect.
d) The statement is incorrect but the reason is correct.

8. Pick out the odd one? Justify your answer. (1)

(Resonance, Reverberation, Echo, Multiple Reflection)

9. Select the characteristics related to Longitudinal wave. (1)

i) Particles in the medium vibrate parallel to the direction of propagation of the wave.
ii) Compressions and Rarefactions are formed.
iii) No Pressure variations occur in the medium.
iv) Crests and troughs are formed.

a) i and iii only b) iii and iv only
c) i, ii and iv only d) i and ii only

10. A longitudinal wave produces 100 Compressions and 100 Rarefactions in one second. What is the period of the wave? (1)

11. Why are the ceilings in large halls curved? (1)

12. Oscillation is the movement of an object in both directions at regular intervals around its equilibrium position.



a) What is the maximum displacement from the equilibrium position to one side in the figure? (1)

(2a, $a/2$, a)

13.



a) What type of wave does the figure indicate? (1)

b) Which of the following indicates the Wavelength of the wave? (1)

(R_1 to R_2, C_1 to R_2,
R_2 to C_3, C_1 to C_3)

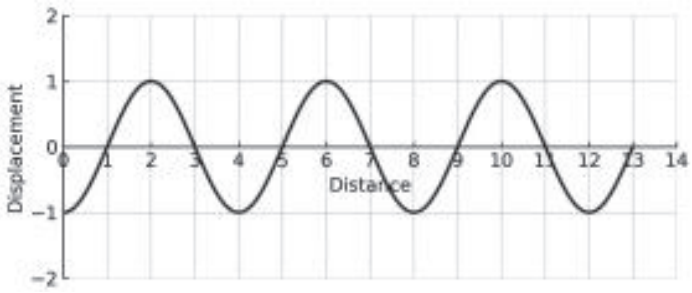
14. How far does travel a wave with a frequency of 2000 Hz and a wavelength of 40 cm in 4 second? (2)

15. Which instrument is used to find the distance to an object under the sea? What are the waves used in this? (2)

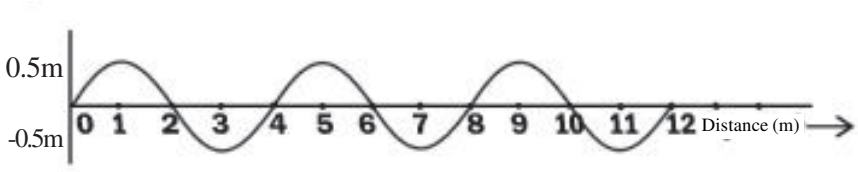
16. The frequency of sound that humans can hear is from 20 Hz to 20 KHz? Then what is the highest wavelength of sound that humans can hear?(Speed of sound=350m/s) (2)

17. How is the image of organs formed in ultrasonography? (2)

18. Graphic representation of a wave traveling at a speed of 3 m/s is given. Find the wave length, period and frequency? 3)



19. Observe the graph. A wave takes 3 second to travel 1200 meter.



a) Find the Amplitude, Wavelength and Frequency of the wave (2)

b) How long does it take for the wave to complete one oscillation? (1)

20. As a result of the earthquake, waves travel through the earth's layers.

a) What is the name of these waves? (1)

b) Which of the following organism can hear these waves? (1)

(Moths, Pigeons, Bats, Cats)

c) Name the scale used to determine the intensity of earthquakes. (1)

21. The different positions of the pendulum when it oscillates are shown in the figure. O is the equilibrium position and A and B are the maximum distances on either side. OA = OB= 6 cm, AB = 12cm.



a) What is its Amplitude? (1)

b) If it takes one second to reach A from O, find its frequency and period? (2)

22. When a child standing between two opposite hills, clapped his hands, an echo was heard from one hill after 0.4 seconds and from the second hill after 0.6 seconds. What is the distance between the hills? (V=350m/s) (3)

23. A sound signal from a ship on the surface of the water hits an object underwater and returns to the ship after 6 seconds. What is the distance travelled by the signal? What is the distance from the water surface to the object?

(The speed of sound in Ocean is considered to be 1500 m/s) (3)

24. Explain the procedure to find the period and frequency of a pendulum using a simple pendulum, meter scale and stopwatch. (3)

25. When sound waves travel through a medium, the particles vibrate. In 3 seconds, the particles complete 60 vibrations and the wave travels 4440 meter. Then,

a) Find the frequency and wave length. (2)

b) Through which of the following mediums could the sound have traveled? (Air, Water, Sea Water) (1)

26. If a simple pendulum of length 10 cm takes 40 s to complete 10 oscillations,

a) What is its period? (1)

b) What is the frequency of another pendulum in resonance with this pendulum? 1)

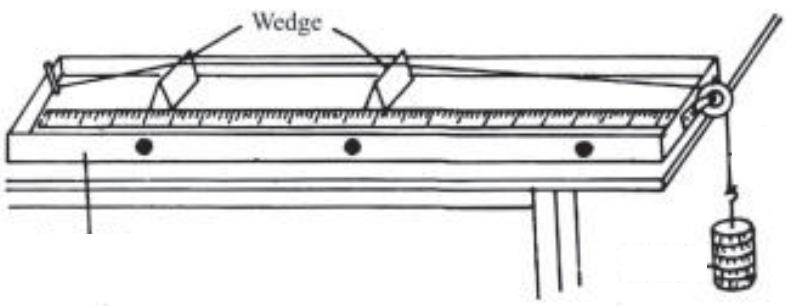
c) What is the length of the second pendulum? (1)

27. There is a table in a classroom with a natural frequency of 520Hz.

a) How many times will the table vibrate in one second if it is tapped on the table? (1)

b) If a tuning fork with a frequency of 256 Hz is excited and its stem is placed on the table, at what frequency will table vibrate? Why? (2)

28.



A Sonometer is shown in the figure. The length of the string is adjusted by placing two pieces of wood. A small piece of paper is folded over the string. When a tuning fork of frequency 256 Hz is placed on the sonometer, the string in the sonometer vibrates.

a) What is the frequency of the string? (1)

b) What is the name of this phenomenon? (1)

c) When the length of the string was adjusted by moving the piece of wood, the piece of paper fell down. What is the reason? (2)

29. In an experiment to find the frequency, a string and a bob of diameter 1cm are given.

a) What should be the length of the string if the length of the pendulum is 60 cm? (1)

b) What will be the change in the frequency if the length of the pendulum is reduced? (1)

c) Will the frequency change if a bob of radius 0.7 cm is used without changing the length of the string? Validate. (2)

30. A longitudinal wave takes 40 seconds to complete 120 vibrations. In this time, the wave travels a distance of 960 meters.

a) Find the frequency, wavelength and speed of the wave. (3)

b) What will be the change in the wavelength if the frequency is increased without changing the speed? (1)

31. The graphical representations of the motion of wave (a) and wave (b) are given.

If the distance from O to A is 120 m and the time taken to travel is 0.3 s,

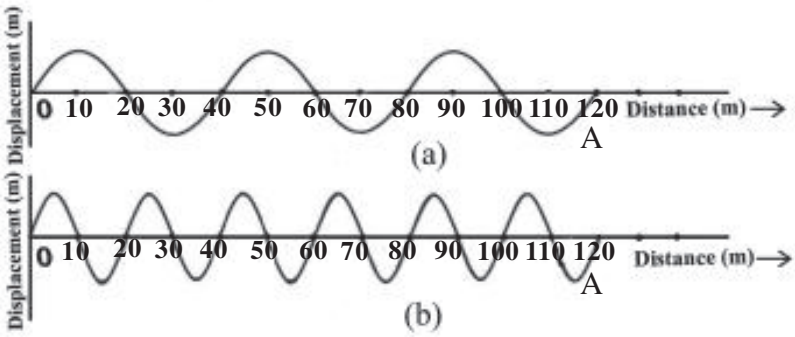


Figure 1

Figure 2

a) Calculate the speed of the waves.

b) Find the frequency of each wave.

c) Which wave has more wavelength?

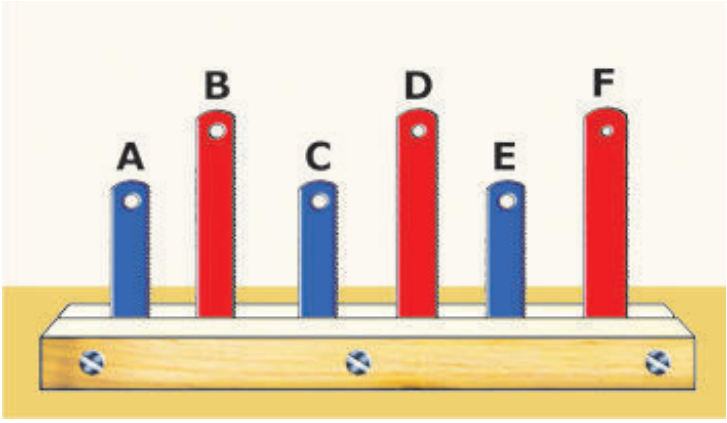
32. If a wave traveling in air at a speed of 350 m/s has a wavelength of 60 m,

a) Can humans hear this sound? Justify your answer.

b) What is this type of sound known as?

c) What natural phenomenon is responsible for the formation of such waves?

33. Observe the picture.



a) If a hacksaw blade A is tapped with a finger, which hacksaw blade/blades will vibrate? (1)

(Only A vibrate, blades A, C and E will vibrate.

blades B, D and F will vibrate, all blades vibrate)

b) Which blades will vibrate at the highest frequency? (1)

c) Write two cases where forced vibration and resonance are used. (2)

34. Earthquakes, volcanic eruptions, massive explosions etc. cause terrible disasters.

a) What is the name of waves formed as a result of this phenomena? (1)

b) What is the scale used to determine the intensity of an earthquake? (1)

c) What measures can be taken to protect against tsunamis? (2)

Chapter 1

SOUND WAVES

Answers

1. d
2. b
3. a
4. a
5. b
6. c
7. c
8. Resonance
   Others are related to the reflection of sound
9. d
10. $\frac{1}{100}$ s
11. It helps to reflect sound from a source and spread it to all parts of the hall
12. a)
13. a) Longitudinal wave
    b) From R1 to R2
14. 3200m
15. Sonar
    Ultrasonic waves
16. 17.5m
17. Ultrasonic waves that travel through body tissues strike and reflect at areas of varying density in the tissues.

    These waves are converted into electric signals to form an image of the organ.
18. Wave length = 4m
    Frequency = 0.75Hz
    Period = 1.33 s

19. a) Amplitude = 0.5m
Wave length = 4m
Frequency = 100Hz
b) $\frac{1}{100}$ s

20. a) Seismic waves
b) Pigeon
c) Richter scale

21. a) 6cm
b) Frequency = $\frac{1}{4}$ Hz , Period = 4s

22. 175m

23. Distance travelled by the signal = 9000m
Distance = 4500m

24. Write the procedure of experiment

25. a) Frequency = 20Hz
Wave length = 74m
b) Water

26. a) 4s
b) 0.25Hz
c) 10 cm

27. a) 520
b) 256Hz
Forced vibration

28. a) 256Hz
b) Forced vibration
c) The freequency varies as the length changes.
When the natural frequency of the string is 256Hz, resonance occurs. The amplitude of the string increases and the piece of paper falls down.

29. a) 59.5cm
b) Increases

c) Varies. The lenth of the pendulam increases as the radius of the bob increases. Then the frequency decreases.

30. a) Frequency = 3Hz
Wave length = 8m
Speed = 24m/s
b) Wave length decreases

31. a) 400m/s
b) 10 Hz, 20 Hz
c) Figure 1

32. a) No. The frequency is less than 20Hz
b) Infrasonic
c) Earth quake

33. a) All waves
b) A, C, E
c) In MRI scanning, for radio tuning

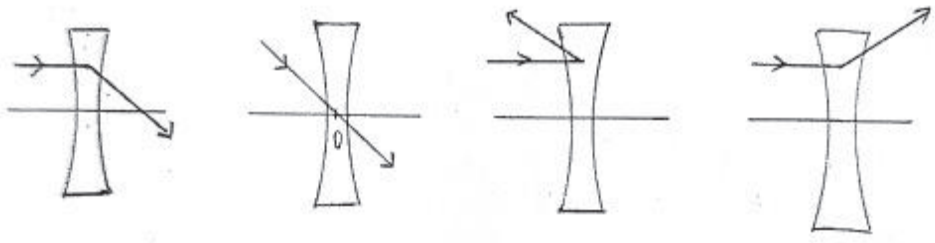
34. a) Seismic vibrates
b) Richter scale
c) Follow the instructions given by the official tsunami warning centres. Do not waste time trying to pick up belongings in the rush to escape, and try to protect yourself.

Chapter 2

LENS

1. Which of the following statements about convex lenses is correct?
   a) A convex lens always forms a real image.
   b) A convex lens always forms a virtual image.
   c) The nature of the image formed by a convex lens depends on the position of the object.
   d) A convex lens cannot form an image on a screen. (1)

2. Which of the following image is correct?



   a) a & b
   b) b & c
   c) c & d
   d) b & d (1)

3. What type of lense should be used to obtain a large and inverted image of an object (1)
   a) Convex
   b) Concave
   c) both can be used
   d) neither of this

4. The focal length of a convex lenses 20 CM which of the following is correct about the image of an object placed 40 cm from this lenes (1)
   a) real and diminished
   b) real and magnified

c) Real and same size as the subject

d) Virtual and same size as the object

5. Statement : The focus of a convex lens is virtual.

Reason : Parallel light rays incident on a convex lens are focused at a point after refraction. (1)

a) Statement and reason are correct; the reason explains the statement

b) Statement and reason are correct; however, the reason does not explain the statement

c) Both the statement and the reason are incorrect

d) Statement is incorrect reason is correct

6. A child tries to burn paper using a lens from a certain distance. After some time, he sees that the paper is burning. Which of the following is the correct reason for the paper burning? (1)

a) Concave lens converges light rays

b) Convex lens converges light

c) Concave lens diverges light

d) Convex lens diverges light

7. Which of the following is suitable for use as an eyepiece and objective of a microscope? (1)

a) The focal length of the eyepiece is greater than that of the objective.

b) The focal length of the objective is greater than that of the eyepiece.

c) The focal length of the eyepiece and objective should be the same.

d) None of the above

8. Statement : When choosing lenses for telescope construction, the focal length and aperture of the objective lens should be increased.

Reason : Increasing the focal length causes the light rays coming from far away to converge, resulting in a smaller, real image.

Increasing the aperture allows more light from distant objects to pass through the lens, resulting in a real image. (1)

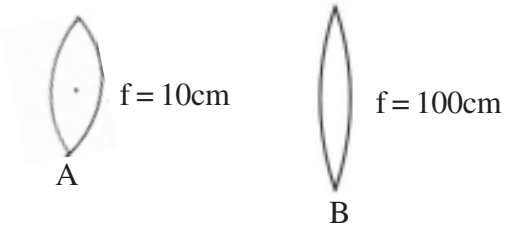
a) Statement and reason are correct; the reason explains the statement

b) Statement and reason are correct; however, the reason does not explain the statement

c) Both the statement and the reason are incorrect

d) Statement is incorrect reason is correct

9. Which of the following lenses can be selected as the objective and eyepiece of a telescope? (1)



a) Eyepiece A, Objective B

b) Objective A, Eyepiece B

c) B can be used as both objective and eyepiece

d) A can be used as both objective and eyepiece

10. What is the focal length of a lens with -2D power? (1)

a) -0.5cm b) -50cm c) -5cm d) -2cm

11. Statement : The focus of a convex lens is real.
Reason : Parallel light rays incident on a convex lens are focused at a point after refraction.

a) Statement and reason are correct; the reason explains the statement

b) Statement and reason are correct; however, the reason does not explain the statement

c) Both the statement and the reason are incorrect

d) Statement is incorrect reason is correct (1)

12. Which of the following dimensions is not possible for a convex lens?

(i) u = +25cm (ii) h_0 = +1cm (iii) f=+2cm (iv) v = +4cm

a) i & iv b) iii only

c) iii & iv d) i,iii & iv (1)

13. Select the correct statements related to a refracting telescope.

i) The focal length and aperture of the objective lens should be greater.

ii) The focal length and aperture of the objective lens should be short.

iii) The image formed by the eyepiece must be real.

iv) The eyepiece lens focal length should be short.

a) i & iv b) iii only

c) ii & iii d) ii only (1)

14. Statement : Increasing the focal length of the objective lens is not beneficial in a compound microscope.

Reason : The greater the focal length of the objective lens, the greater the magnification.

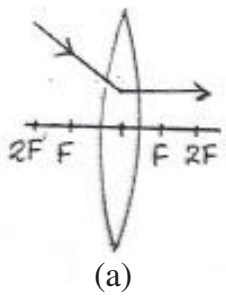
a) Statement and reason are correct; the reason explains the statement

b) Statement and reason are correct; however, the reason does not explain the statement

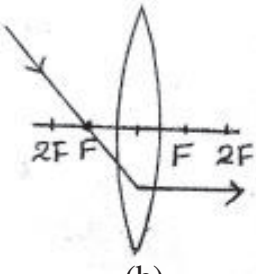
c) Both the statement and the reason are incorrect

d) Statement is correct reason is incorrect (1)

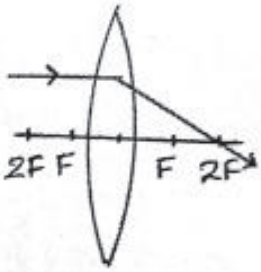
15. Observe the path of a ray of light passing through a convex lens and select the correct image. (1)



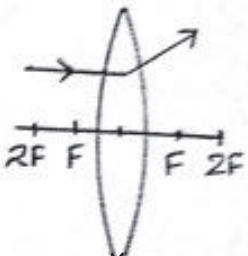
(a)



(b)



(c)



(d)

16. Complete the table

Characteristics of image	Inverted/ Erect	Magnification (Positive/Negative)
Real	(a)	(b). .....
Virtual	(c)	(d) .....

(2)

17. Observe the statements given below and Match the statements with concave lens and convex lens. (2)

a) Always forming a diminished and virtual image

b) Forms a real or virtual image depending on the position of the object

c) A lens that is thick in the middle and converges light

d) A lens that is thin in the middle and diverges light.

18. A camera uses a lens with a focal length of +10 cm.

a) What kind of lens is this? (1)

b) Calculate the power of the lens (1)

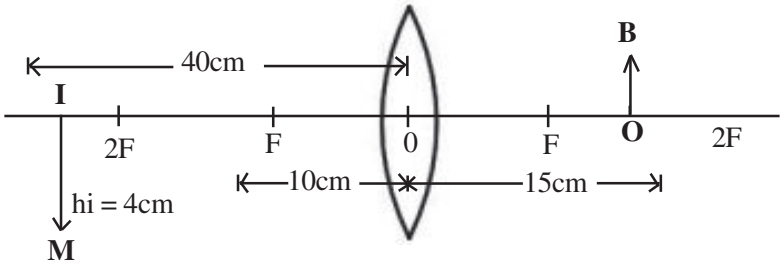
19. A student who was having difficulty reading the letters written on the board went to the doctor and was prescribed a lens with a power of -2D. (1)

a) What kind of lens is this?

b) What is the focal length of the lens?

20. How far should the object be from the lens if an image of 6 cm high of an object 2 cm high is to be formed on a screen 1.5 m away? (2)

21. The following diagram is analysed using the New Cartesian sign convention method. The observations written by a child are given below:



u = -15 cm, v = -40 cm, f = -10 cm, h_i = -4 cm.

Find the errors in these observations and explain the reasons. (2)

22. The ophthalmologist prescribed a lens with a power of +4 D. (2)

a) What kind of lens is this?

b) calculate the focal length of the lens.

23. The magnification of a convex lens is given as -0.25.

a) Write the characteristics of the image. (1)

b) The height of the image is how many times the height of the object? (1)

24. Draw the ray diagram of the image formation of Reading lens. (2)

25. A reading lens is used to magnify objects. Then what is the purpose of a microscope? Explain.

26. The focal length of a convex lens is 20 cm. When an object is placed in front of it, a real and inverted image of the same size as the object is formed.

a) How far is the object from the lens? (1)

b) Is it possible to form this image on the screen? Why? (1)

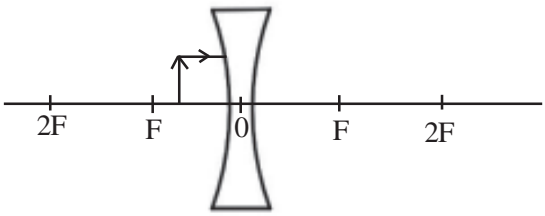
c) Calculate the magnification. (1)

27. A lens is a transparent medium in which each refracting surface is part of the spheres

a) What is the optical phenomenon that occurs when light passes through a lens? (1)

b) How does the principal focus of convex and concave lenses differ? (2)

28 An incomplete diagram related to the image formation of a concave lens is given.



a) Complete the picture and find the position of the image. (2)

b) Write the characteristics of the image formed if a convex lens were used instead of a concave lens in the given image. (1)

29. As part of a science project, the children decided to build a refracting telescope.

a) What are the optical phenomena that can be exploited in telescopes? (1)

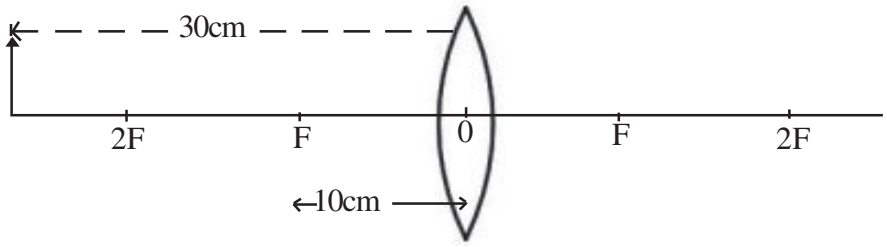
b) What are the key considerations and factors to keep in mind when building a telescope? (2)

30. The focal length of a lens is 10 cm. If an object is placed at a distance of 15 cm,

a) How far should the screen be placed to get a real image? (1)

b) If the height of the image formed on the screen is 2 cm, what is the height of the object? (2)

31. Analyse the picture and answer the questions given below.



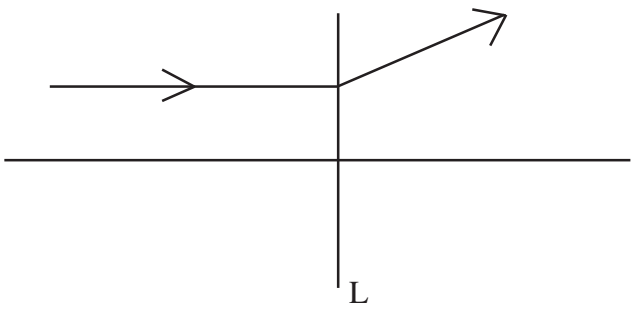
a) Complete the diagram. (1)

b) Calculate the distance to the image. (1)

c) Calculate the magnification. (1)

d) If the height of the image is 4 cm, what is the height of the object? (1)

32. Observe the diagram given below.

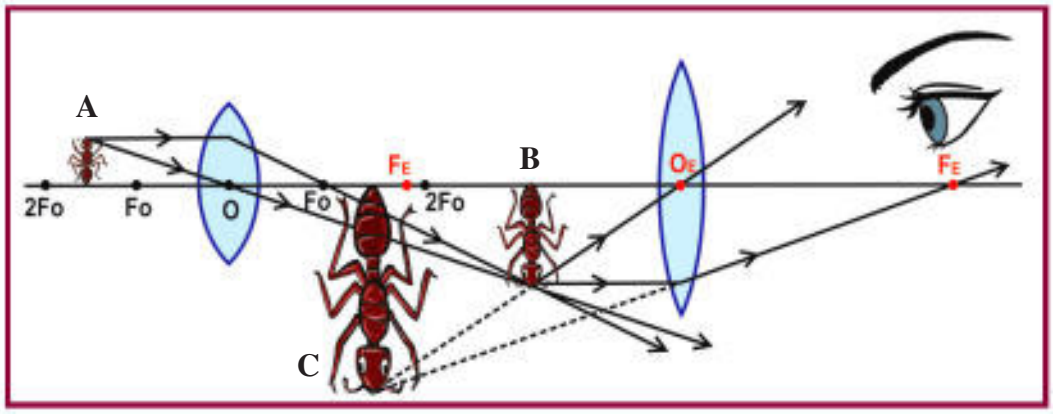


a) What type of lens does L stand for? (1)

b) Complete the image formation picture. (2)

c) Write the characteristics of the image. (1)

33. Observe the image formation of the microscope.



a) What is the object related to eyepiece? (1)

i) A ii) B iii) C

b) Where should the object be located with respect to the eyepiece? (1)

i) at F

ii) between F and 2F

iii) between F and optical centre of eyepiece

iv) at 2F (1)

c) Why is it not advisable to increase the focal length of the objective lens?(2)

34. An object 1 cm high is placed 20 cm away from a lens with a power of -5D.

a) What is the distance to the image? (1)

b) Write the characteristics of the image. (1)

c) Calculate the magnification (1)

d) Is it possible to form an image on the screen? Why? (1)

35. The focal length of a convex lens is 20 cm. An object 2 cm high is placed 40 cm from the optical centre of this lens.

a) Calculate the magnification? (1)

b) Draw the ray diagram of the image formation (2)

c) What are the characteristics of the image obtained? (1)

36. An object 30 cm away from a concave lens produces an image one-third the size of the object.

a) What are the characteristics of the image obtained? (1)

b) How far will the image be from the lens? (1)

c) find the focal length of the lens (2)

37. A,B,C are three convex lenses. Their focal lengths and the distance to the object placed in front of them are given in the table below.

lens	Focal length(cm)	Distance to the object(m)
A	20	45
B	15	30
C	30	20

a) Which of the given lenses creates a diminished image of the object?

b) In which lens will a virtual image of an object be formed?

c) Draw a diagram of the image formation of lens B and write two characteristics of the image.

38. There are four lenses, A, B, C, and D, and their characteristics are given in the table below.

Lens	Convex/Concave	Focal length
A	Convex	15cm
B	Concave	20cm
C	Convex	45cm
D	Concave	45cm

a) Which of these lenses should be used as the eyepiece of a compound micro scope?

b) Which of these lenses should be used as the objective of a refracting telescope?

c) Justify your answers.

Chapter 2

LENS

Answers

1. b
2. d
3. a
4. c
5. a
6. b
7. a
8. b
9. a
10. b
11. a
12. d
13. a
14. d
15. b

16. a) Inverted
    b) Negative
    c) Erect
    d) Positive

17. a) Concave
    b) Convex
    c) Convex
    d) Concave

18. a) Convex
    b) f = 1cm = 0.1m

$$P = \frac{1}{f} = \frac{1}{0.1} = +10D$$

19. a) Concave

b) $f = 1/P = 1/-2$

$= -0.5m$

20. $h_0 = +2cm$ $h_i = -6cm$

V = 1.5m = 150cm

$h_i/h_0 = V/u$

$u = \frac{h_0 x V}{h_i} = \frac{(+2)x(+150)}{-6}$

= -50cm

21. The wrong observations are V = -40cm, f = -10cm

According to cartesian sign convention distance measured in the same direction as the incident ray should be positive.

22. a) Convex

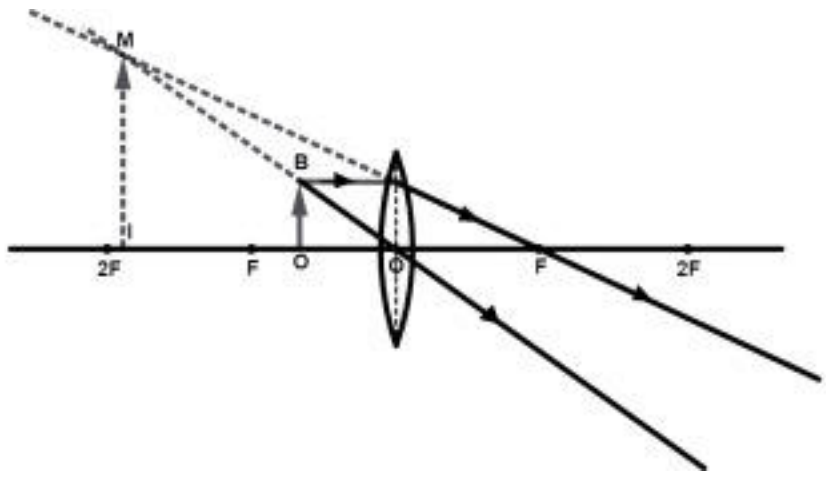
b) $f = 1/p = 1/+4$

= 0.25m = +25cm

23. a) Inverted and Diminished

b) $1/4$ times

24.



25. In reading lens ,when the object is kept in between optic centre and focus we get enlarged erect image.

In a microscope, enlarged image obtained by the objective lens is again magnified by the eyepiece.

26. a) u = - 40cm

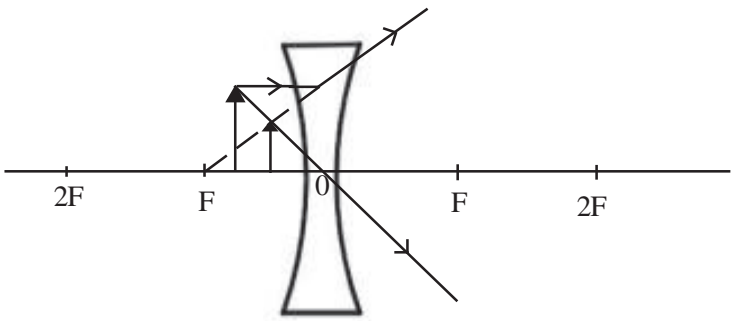
b) Yes. Becouse the image formed is real.

c) -1

27. a) Refraction

b) Light rays near and parallel to the optic axis incident on a convex lens,after refraction converges at a point on the optic axis on the other side of the lens. This is the principle focus of convex lens. It should be real. The light rays near and parallel to the optic axis incident on a concave lens,a fter refraction diverges from a point on the optic axis on the same side of the lens. This is the principle focus of concave lens. It should be virtual.

28. a)



b) Erect and Magnified

29. a) Reflection and Refraction

b)
- The focal length and aperture of the objective lens should be greater.
- The focal length and aperture of the eyepies lens should be lesser.
- Use high quality lenses.

30. a) f = +10cm u = -15cm v=?

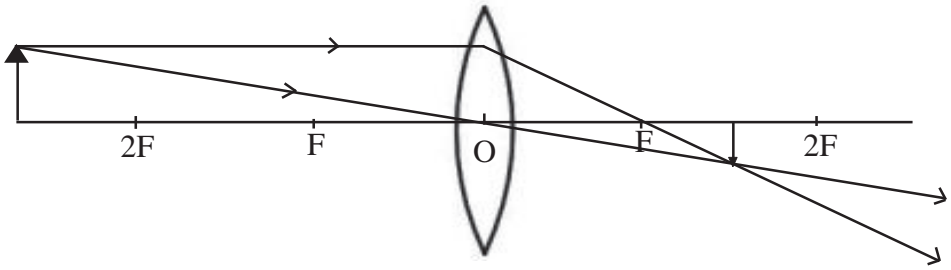
v = uf / (u+f)

$$v = \frac{-15x+10}{-15+10} = +30cm$$

b) h_i = -2cm

$$h_i / h_0 = v / u = \frac{-15x-2}{+30} = +1cm$$

31. a)



b) u = -30cm f = +10cm

$$v = \frac{uf}{u+f} = \frac{-30x10}{-30+10} = \frac{-300}{-20} = +15cm$$

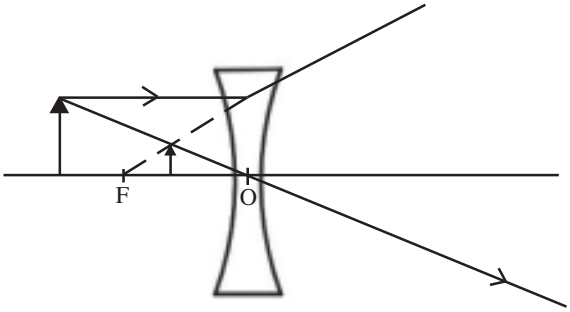
c) $m = v/u = +15/-30 = -0.5$

d) $h_i / h_o = m$

$$h_O = \frac{h_i}{m} = \frac{-4}{-0.5} = 4x2 = 8cm$$

32. a) Concave

b)



c) Erect and Diminished

33. a) B

b) iii

c) If the focal length of the objective lens is longer,the size of the image will be smaller. This means magnification will be lesser.

34. a) P = -5 D, u = -20 cm, h_o = 1 cm

$$f = 1/-5 = -0.2,\ m = -20\ cm$$

$$v = \frac{uf}{u+f} = \frac{400}{-400} = -10\ cm$$

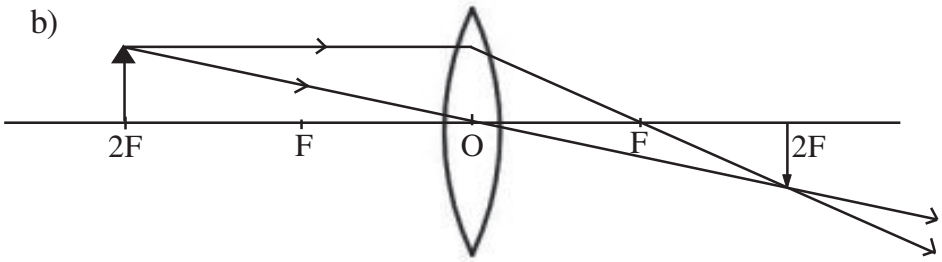
b) Virtual and Erect

c) $$m = \left(v/u\right) = \frac{-10}{-20} = -\frac{1}{2}$$

d) No. because the image formed is virtual

35. a) $$m = \frac{v}{u} = 40/-40 = -1$$

b)



c) Inverted, Real and Same size

36. a) Virtual, Erect and Diminished

b) $$m = +\frac{1}{3}$$

$$\frac{v}{u} = m$$

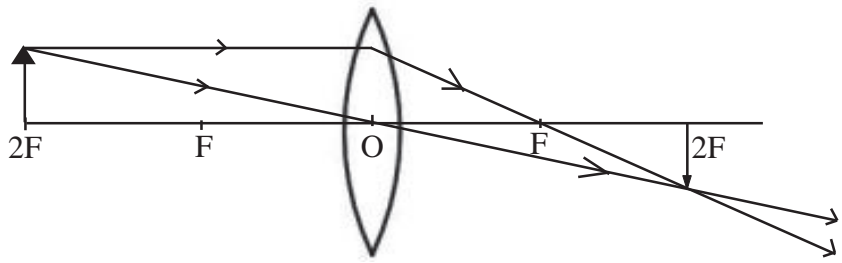
$$v = mxu = 1/3\ x - 30 = -10cm$$

c) $f = \frac{uv}{u-v} = \frac{300}{-20} = -15cm$

37. a) A

b) C

c)



Real, Inverted, Same size

38. a) C

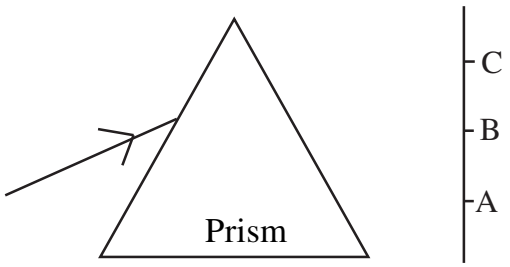
b) C

c) To get a magnified and virtual image of its object,focal length of the eyepiece in a microscope should be large. In a telescope ,the objective forms a small ,real and inverted image of a distant object. Thus we use convex lens of greater length used as objective of telescope.

Chapter 3

The World of Colours and Vision

1.



A, B, C are some of the coloures appeared on the screen after the dispersion of light. Which of the following is correct? (1)

i) A - A-Red B - Blue C - Violet

ii) A - Violet B - Green C - Yellow

iii) A - Violet B - Red C - Green

iv) A - Green B - Yellow C - A-Red

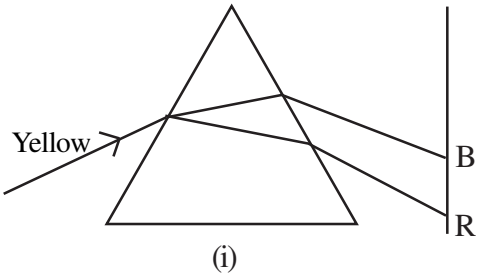
a) Both i) and iv) are correct.

b) Both i) and ii) are correct.

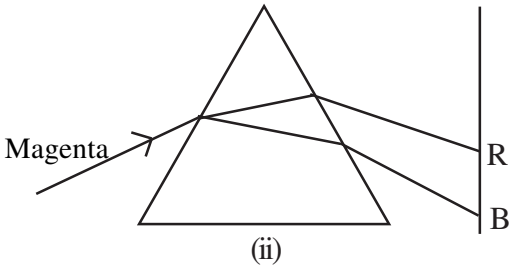
c) Both i) and iii) are correct.

d) Both ii) and iv) are correct.

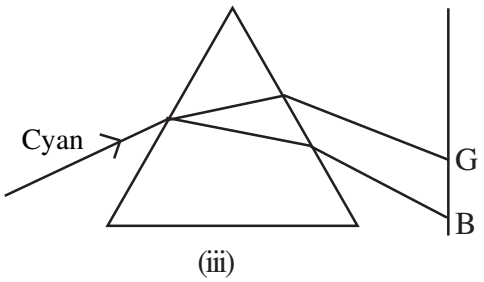
2. The colours obtained on the screen when different coloured light is passed through various prisms are given.



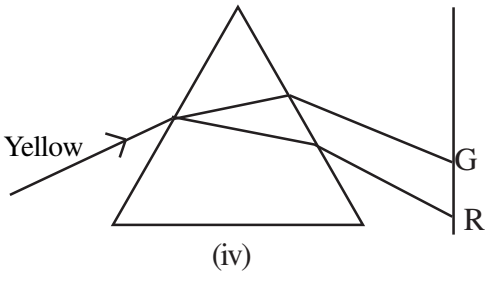
(i)



(ii)



(iii)



(iv)

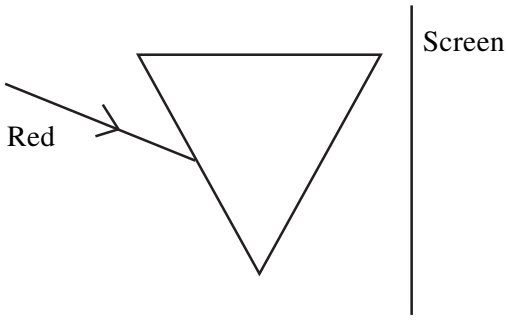
a) Both i and ii are correct.

b) Both i and iv are false.

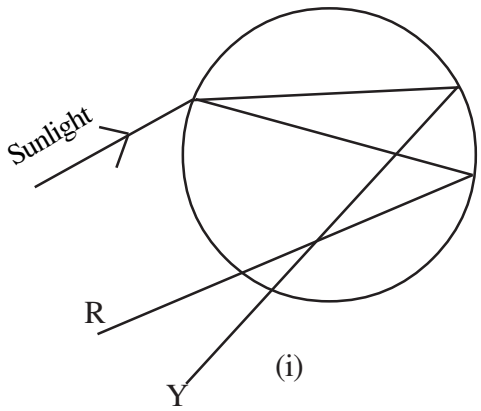
c) Both i and iii are correct.

d) Both ii and iv are correct. (1)

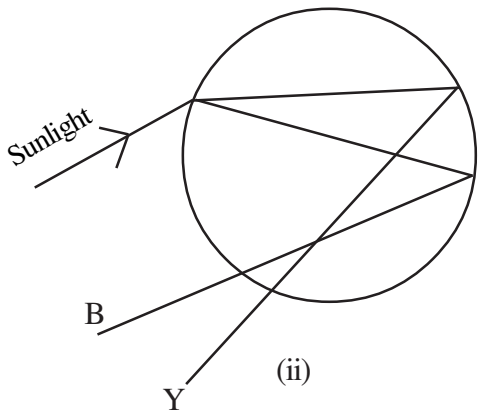
3. Red light falling on a prism is shown. Complete the picture (1)



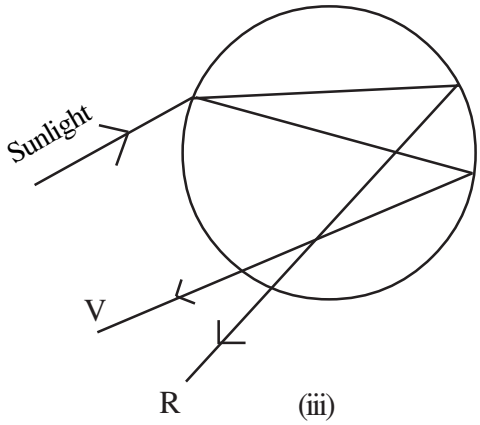
4. Various instances of sunlight falling on water molecule and undergoing dispersion are depicted.



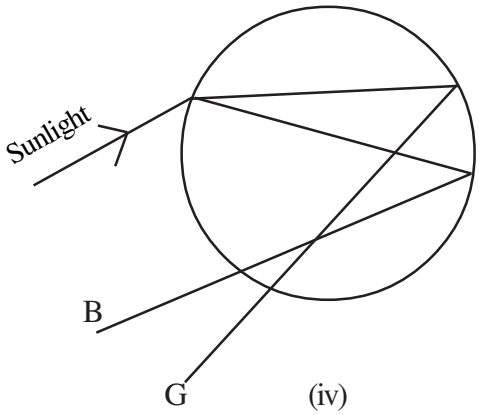
(i)



(ii)



(iii)



(iv)

Observe the figure and choose the correct option.

b) Both ii and iii are correct.

c) Both i and iv are incorrect.

d) Both ii and iii are correct. (1)

5. The rays in the electromagnetic spectrum are given.

Select the one written in the increasing order of wave length.

i) Gamma, infrared, visible light, radio

ii) Radio, infrared, X-ray, gamma

iii) X-ray, gamma, infrared, radio

iv) Ultraviolet, visible light, infrared, radio

a) Both i) and ii) are correct.

b) ii only correct.

c) iv) only correct.

d) Both iii & iv are incorrect

6. Statement: Longer wavelengths have less scattering.

Choose the one that best describes the statement from the following

i) vehicle brake lights are red in colour

ii) The sky becomes blue

iii) The horizon is red at sunrise and sunset

iv) The ocean turns blue

a) i), ii)

b) i), iii)

c) iii), iv)

d) i), iv) (1)

7. When light was incident on a magenta object, the object appeared blue.

Which of the following is the light incident on the object?

i) magenta

ii) red or yellow

iii) cyan or blue

iv) blue

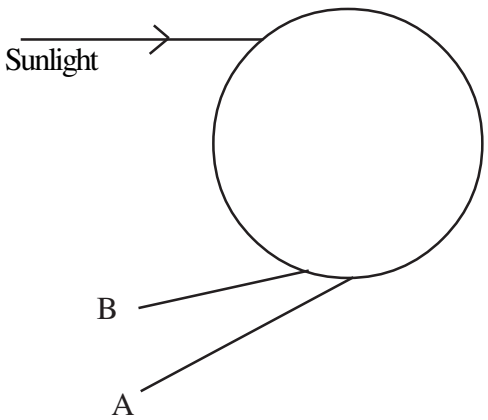
a) Both i & iii are correct

b) Both iii & iv are correct

c) Both ii & iv are correct

d) Both ii & iii are correct (1)

8. A rainbow is formed in the atmosphere when light passes through a water droplet. In this case, some of the colours emitted from the water droplet are A and B.



i) A - Red B-Violet

ii) A-Indigo B- Yellow

iii) A- Blue B- Violet

iv) A- Green B-Red

a) Both i & ii are correct

b) Both iii & iv are correct

c) Both i & iii are correct

d) Both ii & iv are correct (1)

9. If sunlight is passed through a prism, the light rays can be splitted into their constituent colours. Which colour deviates the least? (1)

a) Red b) Green c) Blue d) Violet

10. Statement : The colours red, green, and blue are called primary colours.

Reason : Not only white light, but all other colours can be produced using red, green, and blue.

Choose the correct one from the following. (1)

a) The statement and the reason are correct. reason is the correct explanation for the statement

b) The statement and the reason are correct. The reason is not the correct explanation for the statement

c) The statement and the reason are incorrect

d) The statement is false, but the reason is true

11. A student burned paper using a lens. Which of the following is the reason for the paper burning?

a) Gamma radiation b) X-rays

c) Ultraviolet radiation d) Infrared radiation (1)

12. What are the colour pairs that can produce white light?

a) red-yellow b) red-green

c) red-blue d) red-cyan (1)

13. Statement : a burning stick is rotated very fast,a ring of fire is seen.

Reason : When we quickly remove an object from our field of vision the visual experience of that object persist for about $\frac{1}{16}$ of a second.

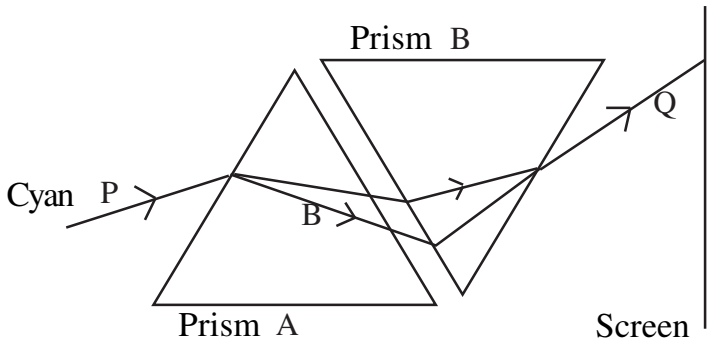
a) Statement and reason are correct; the reason explains the statement

b) Statement and reason are correct; however, the reason does not explain the statement

c) Both the statement and the reason are incorrect

d) Statement is incorrect reason is correct (1)

14. A beam of composite light is passed through prisms A and B as shown in the figure and projected onto a screen.



a) Which light is incident on the screen at point Q? (1)

b) Which phenomenon takes here? (1)

15. Composite light is light that is produced by combining different colours. Write an experimental procedure to prove that composite light is produced by combining different colours.

16. The observations made when different coloured light was shone on opaque objects of different colours are recorded in the table.

Complete the table. (2)

The colour of the object	Incident light	Colour of the object seen
Yellow	blue	a) .....
Magenta	b)	red

17. Explain why a saree that appeared light red under the light of a fluorescent tube in a cloth shop appeared dark red in daylight.

18. The information given is related to two radiations in the electromagnetic spectrum. Correct it if there is any error.

	A	B
1. Radio waves	High frequency	Longer wavelength
2. Gamma rays	Low frequency	Shorter wavelength

(2)

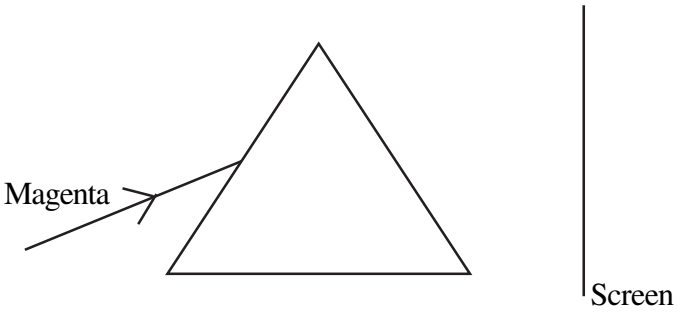
19. The child can see a nearby person's image clearly. When the person moves away, the child can still see clearly.

a) What is the change in the power of the child's eye lens in this situation? (1)

b) How can the eye change its lens power? (1)

20. What is light pollution? Write two disadvantages caused by it. (2)

21. Magenta light incident on the surface as shown in the picture.



a) Complete the figure. (1)

b) What is the phenomenon happening here? Explain. (1)

22. Name the electromagnetic radiation with a lesser wave length that is close to visible light. Write any one advantage of this? (2)

23. What colour can the coloured objects be seen as, based on the lighting in the room? (2)

Colour of the object	The light in the room	
	Green	Red
Cyan	(a)	(b)
Red	(c)	(d)

24. In what colour can the following coloured objects be seen in white light through the given colour filter? (2)

Colour of the object	Colour of filter	
	green	cyan
Magenta	(a)	(b)
Blue	(c)	(d)

25. Some information related to eye defects is given in the table. Check the connection and correct if there is any error. (2)

	A	B
1. Short sightedness	Far point at infinity	Size of eyeball is big
2. Long sightedness	Near point more than 25cm	Focal length of the lens is short

26. Some information related to eye defects is given in the table. Check the connection and correct if there is any error. (2)

	A	B
1) Short sightedness	Size of eyeball is big	The power of the lens is less
2) Long sightedness	The power of the lens is less	Can be rectified using a concave lens with suitable power

27. Complete the table

Secondary Colour	Complementary colour
yellow	a) .....
b)	red

(2)

28. Sunlight falls on the moon in the same way it falls on the earth, but the sky on the earth is blue and the sky on the moon is dark. Explain Why?

29. The child can clearly see a person standing on the ground, whether they are very far away or close.

a) What is the change in the curvature of the child's eye lens when the person in the distance comes closer? What is the change in power? (1)

b) What is the name given to the eye's ability to make this change? How does the eye make this function possible? (1)

30. Scattering is the irregular and partial directional deviation of light when it encounters particles in a medium.

a) Write the factor influencing scattering? (1)

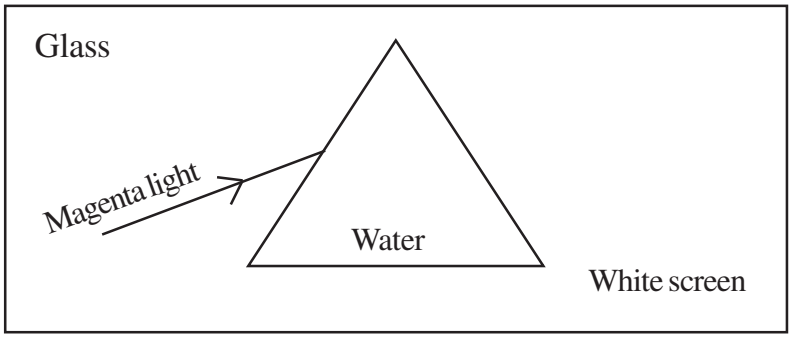
b) Red light can travel a long distance in the atmosphere. Explain Why? (1)

31. To obtain a clear image, images of both distant and near objects must be formed on the retina.

a) What is the name given to the eye's ability to form an image on the retina? (1)

b) How are images of objects at different distances formed on the retina? (1)

32. In the picture, a square glass pillar has a prism-shaped space filled with water in the middle. A magenta light is incident on the prism



a) Complete the figure. (2)

b) What phenomenon is happening here? Explain. (1)

33. If the seven colours of the spectrum are painted on a Newton's colour disk in equal amounts and with equal intensity, and the disk is rotated rapidly, it appears white.

a) What is the reason for this phenomenon? (1)

b) How many new secondary colours can be made using the primary colours in the colour wheel? Name them? (1)

c) What colours should be painted in equal amounts and at equal intensity to make the disk appear white using the fewest colours? (1)

34. Classify the following as primary and secondary colours.

Red, Yellow, Green, Magenta

a)

Primary Colours	Secondary Colours

(2)

b) Which of the following colours should be used to produce white light using the smallest number of colours? (1)

35. The electromagnetic spectrum is given.

P VIBGYOR **Q**

a) Identify and write down the types of radiation P and Q. (1)

b) Which of these radiations has the higher frequency? (1)

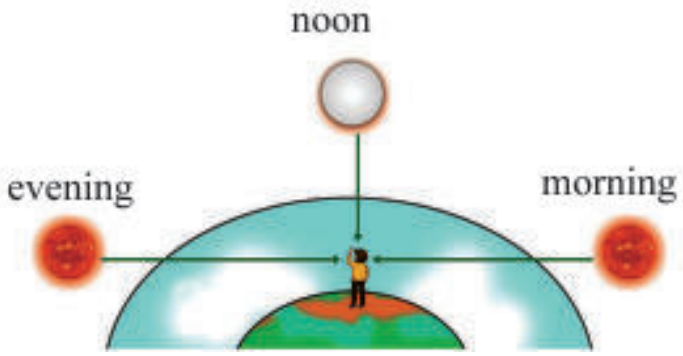
c) State one use of Radiation 'P'? (1)

36. A disk painted green and blue is spinning very fast.

a) What colour will the disk appear in sunlight? (1)

b) What would be the observation if this activity was performed in red light? Justify your answer. (2)

37. The observations of a child when the sun is at different positions are depicted.



a) In which cases does the sun appear white? Justify your answer. (2)

b) The deep sea appears blue. Why? (1)

38. When dil HCl was added to sodium thiosulphate solution and white light was passed through it using a torch and observed for some time, the solution turned blue and the screen turned red.

a) What is the reason for this phenomenon? (1)

b) The next day, when light was paused through the same medium. Why did the screen appear dark and the solution appear white? (1)

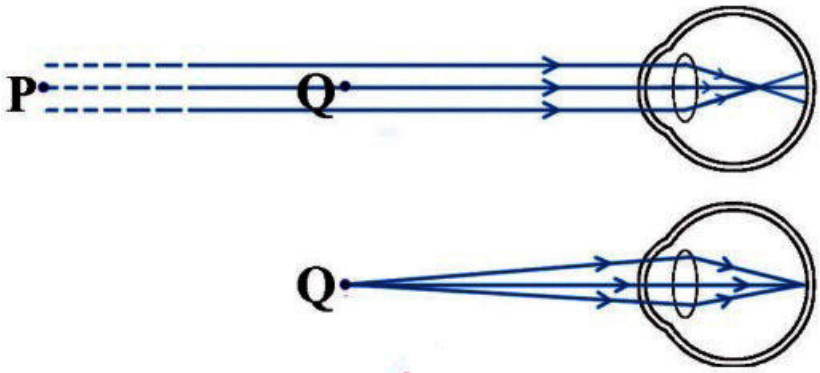
c) Why is a red light used as a danger signal? (1)

39. A candle is placed 20 cm away from a convex lens of focal length 10cm.

a) At what distance should the screen be placed to obtain a clear image?

b) When the position of the object was increased to 30cm. The image became indistinct. What changes should be made to obtain clear image without changing the position of the screen.

40. The images formed in a person's eye in two different situations are shown.



a) What is the name of this eye problem? (1)

b) What is the solution for it? (1)

c) Write any two reasons for this defect? (1)

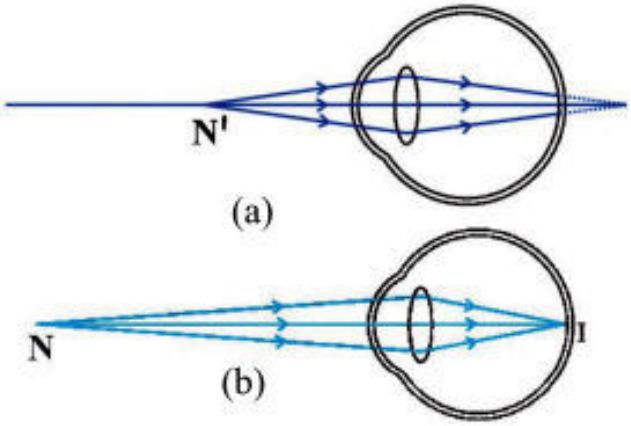
41. An eye doctor suggested using a -2D power lens to correct a person's eye defect.

a) What type of lens is this? What is the focal length of this lens? (1)

b) What kind of eye problem does this person have? (1)

c) Write the reasons for this defect. (1)

42. A diagram of a person viewing an object at two different distances is given.



a) Which limitation of the eye lens is responsible for this defect? Write one more reason for this defect. (2)

b) How can this be corrected? (1)

43. When older people read, they often hold the book at a distance.

a) What is this visual impairment called? (1)

b) What is the reason for this? (1)

c) Write a solution for this. (1)

44. Mahesh, standing in the school playground, saw a beautiful rainbow in the western direction.

a) When did Mahesh see the rainbow?

(Morning, Afternoon, Evening) (1)

b) Draw a diagram showing the separation of colours that occurs in a water droplet when a rainbow is formed. (1)

c) Explain how a rainbow is formed. (1)

45. Explain with reasons.

i) The path of light passing through the suspension can be clearly seen. (1)

ii) The sky usually appears blue (1)

iii) The colour of the sun at sunset appears red, yellow or orange. (1)

46. When Newton's colour disk rotates rapidly, white appears.

a) What phenomenon causes this? (1)

b) Explain this phenomenon? (1)

c) Write another example of this phenomenon? (1)

47. The sky appears blue due to scattering.

a) What is meant by scattering? (1)

b) How is the above phenomenon related to the Tyndall effect? (1)

c) What colour does the sun appear when viewed from the moon? Explain. (1)

48. Solar thermal plants use the thermal energy in solar radiation to generate energy.

a) What radiation causes heat? (1)

b) Write two situations in which this radiation is useful. (2)

c) Arrange this radiation along with the component colours of visible light in ascending order of wavelength. (1)

49. When an object is 50cm away from the eye, a clear image is formed on the retina, which is 3cm behind the lens of the eye.

a) Identify this person's eye defect. (1)

b) Which type of lens is used to correct this defect? (1)

c) Write any two reasons for this defect? (2)

50. A disk painted with the colours red and green in equal amounts and intensities is rotated rapidly and observed in sunlight.

a) What will be the observation? Justify your answer. (2)

b) What would be the observation if the same experiment were performed in blue light instead of sunlight? (2)

51. When you shine a torch into the sky on a snowy night, a trail of light appears.

a) Explain the phenomenon that caused this. (2)

b) When dilute hydrochloric acid was added to a sodium thiosulphate solution and light was passed through it, the solution appeared blue. But why doesn't the atmosphere diffuse the blue colour when light passes through snow? (2)

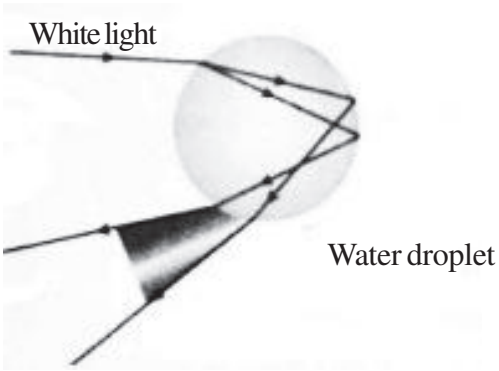
52. The near point of a person with long sightedness is 50 cm.

a) Can this person see objects 90cm away? Why? (2)

b) Write the reasons for this person's eye defect. (1)

c) How can this defect be corrected? (1)

53. Observe the picture and answer the questions given below.



a) How many times does a light ray undergo refraction when entering and exiting a water droplet?

b) How many times does it undergo total internal reflection?

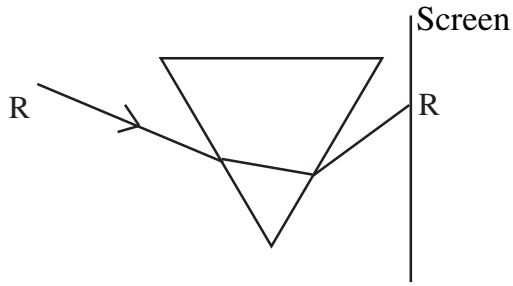
c) What colour is seen in the outer layer of the rainbow?

d) What is the colour of the inner layer of the rainbow?

Chapter 3

The World of Colours and Vision

ANSWERS

1. d) Both ii) and iv) are correct.
2. b) Both i and iv are false.
3. 

4. b) Both ii and iii are correct.
5. c) iv) only correct.
6. b) i), iii)
7. b) Both iii & iv are correct
8. c) Both i & iii are correct
9. b) The statement and the reason are correct. The reason is not the correct explanation for the statement
10. a) The statement and the reason are correct. reason is the correct explanation for the statement
11. d) Infrared radiation
12. d) red-cyan
13. a) Statement and reason are correct; the reason explains the statement
14. a) Cyan

b) Recombination of colours

15. Newton's colour disc or recombination of colours through prism

16. a) Dark
    b) Red

17. The colour spectrum of the light emitted by the fluorescent tube contains less of the longer wavelength colours, so the red colour of the saree is reflected in a smaller amount, making it appear a lighter red.

18. The information given is related to two radiations in the electromagnetic spectrum. Please correct me if there is any error.

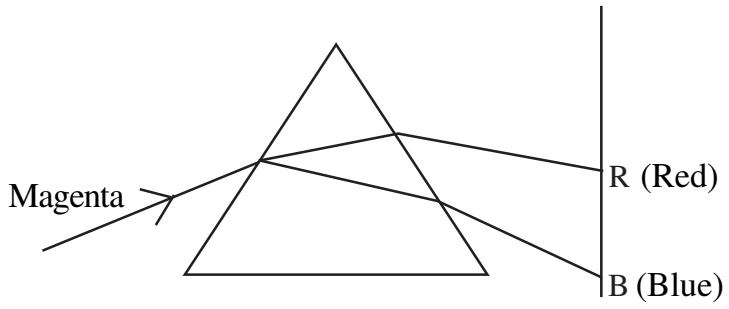
	A	B
1. Radio waves	low frequency	Longer wavelength
2. Gamma rays	high frequency	Shorter wavelength

(2)

19. a) Power reduces
    b) Power of accommodation

20. Light pollution refers to the creation of artificial light in excessive amounts and intensity
    * Causes difficulty during night drive.
    * Makes astronomical observations difficult by obstructing the night sky.
    *The light from multi-storeyed flats misleads migratory birds.

21. a)



b) Dispersion of light is the phenomenon of splitting up of a composite light into its component colours.

22. Ultraviolet radiations
Uses:helps to produce vitamin D in our body (2)

23. a) Green
b) Dark
c) Dark
d) Red

24. a) Dark
b) Blue
c) Dark
d) Blue

25. Some information related to eye defects is given in the table. Check the connection and correct if there is any error. (2)

		A	B
1.	Short sightedness	Far point not at infinity	Size of eyeball is big
2.	Long sightedness	Near point more than 25cm	Focal length of the lens is more

26.

	A	B
1) Short sightedness	Size of eyeball is big	The power of lens is more
2) Long sightedness	The power of the lens is less	Can be rectified using a convex lens with suitable power

27. a) Blue

b) Cyan

28. This is because the Earth's atmosphere scatters sunlight, making the sky appear blue. The Moon has no atmosphere, so sunlight isn't scattered, resulting in a dark sky.

29. a) Curvature of the lens increases,power increases

b) Power of accommodation -The ability of the eye to change the curvature ofthe lens and adjust the focal length so that the image of the object always falls on the retina, regardless of the position of the object, is the power of accommodation of the eye.

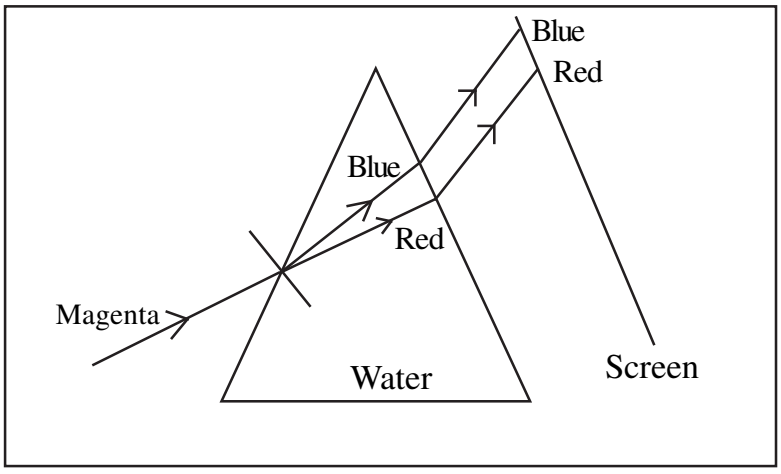
30. a) Size of the particles ,wavelength of light

b) The scattering of red, having a relatively longer wavelength, is very low. Hence red can travel a longer distance through the atmosphere

31. a) Power of accommodation

b) This is achieved by adjusting the curvature of the eye's lens and varying its focal length with the help of the ciliary muscles in the eye. When ciliary muscles contract, the curvature increases and the focal length decreases.

32. a)



b) Dispersion of light, Refraction

33. a) Persistence of vision

b) Magenta, Cyan (Yellow already exist in spectrum)

c) Yellow + Blue

34. a) Primary colours-red, Green

Secondary colours - yellow, Magenta

b) Magenta + Green > White

35. a) P-ultraviolet , Q-infrared

b) P

c) Ultraviolet radiation helps to produce vitamin D in our body.

36. a) Cyan

b) Appears in dark colour. Opaque objects like green and blue absorb red light

37. a) At noon, sunlight only has to travel a short distance through the Earth's atmo sphere, so there is less scattering of sunlight, so all the components of white light are maximally absorbed by the Earth's atmosphere

b) Due to scattering by particles in water, colours with generally shorter wavelengths, such as violet, indigo, and blue, are scattered into the water, resulting in a blue colour.

38. a) Scattering of light

b) When particle size exceeds the wavelength of light, all colors are scattered equally, causing the solution to appear white. Consequently, the screen remains dark because no light reaches it.

c) The scattering of red, having a relatively longer wavelength, is very low. Hence red can travel a longer distance through the atmosphere.

39. a) 20 cm

b) Use a convex lens of focal length 12 cm.

40. a) Short sightedness

b) Use concave lens of suitable power

c) Power of lens is more,size of eyeball is large

41. a) concave lens,

f = -50 cm

b) Short sightedness

c) power of lens is more,size of eyeball is large

42. a) Power of lens is less,size of the eyeball is less.

b) use convex lens with suitable power

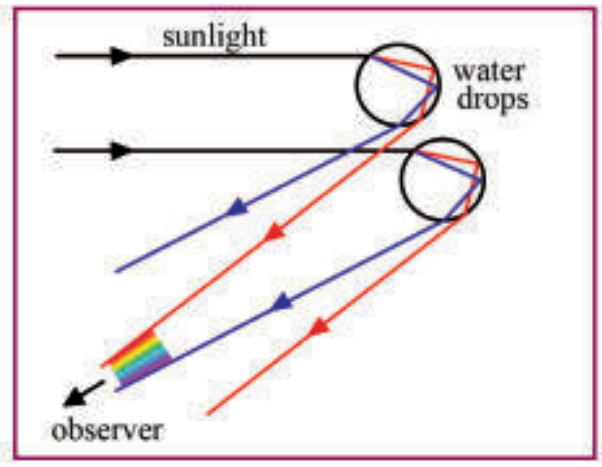
43. a) Presbyopia

b) Efficiency of the ciliary muscles decreases.

c) Use convex lens with suitable power

44. a) Morning

b)



c) A ray of sunlight passing through a water droplet undergoes refraction twice and internal reflection once. This is a natural phenomenon. A rainbow is formed as a result of the combined effect of refraction, dispersion, and internal reflection.

45. i) In a suspension the particles are large enough to scatter light sideways, so the beam becomes visible as a bright line.

ii) When sunlight travels through the atmosphere to reach the Earth, some of its components undergo scattering as it passes through the air. Scattering occurs most for colours with shorter wavelengths such as violet, indigo and blue. This scattered light spreads in the sky. The resultant scattered light that reaches the observer's eye gives the effect of the blue colour. So the sky appears blue.

iii) At sunrise or sunset the light traverses a long atmospheric path, so most blue is scattered out and only the longer redyellow wavelengths reach the eye, giving the sun its reddish hue.

46. a) Persistence of vision

b) When we quickly remove an object from our field of vision the visual experience of that object persists for about 16 1 of a second. This phenomenon is the persistence of vision.

c) When a torch or a burning stick is rotated very fast, a ring of fire is seen.

47. a) Scattering is the irregular and partial directional deviation of light when it encounters particles in a medium.

b) When light rays pass through a colloidal liquid or suspension, they get scattered, causing tiny particles to become illuminated, making the path of light visible. This phenomenon is the Tyndall effect.

c) White.

The Moon has no atmosphere, so sunlight isn't scattered.

48. a) Infrared radiations

b) Remort controller

infrared thermometer

c) VIBGYOR, infrared
Violet, Indigo, Blue, Green, Yellow, Orange, Red, Infrared

49. a) Long sightedness
b) Convex lens
c) Size of eyeball is smaller,power of lens is less

50. a) Yellow
red+green->yellow
b) appears in dark.composite colours reflects only its component colour.blue is a primary colour.

51. a) Tyndall effect
b) The intensity of scattering increases as the size of the colloidal particles increases. Since the size of the snow particles is large, all colors are scattered equally.

52. a) Yes.Some people can see distant objects clearly but cannot see nearby objects clearly. This defect of the eye is long-sightedness.
b) Size of eyeball small,power of the lens is less
c) Using covex lens with suitable power

53. a) Two time
b) One time
c) Red
d) Violet

Chapter 4

MAGNETIC EFFECT OF ELECTRIC CURRENT

1. Observe the statements about the bar magnet given below.

   i) The direction of magnetic flux lines outside the magnet is from the north pole to the south pole.

   ii) The direction of magnetic flux lines inside the magnet is from the north pole to the south pole.

   iii) The direction of magnetic flux lines outside the magnet is from the south pole to the north pole.

   iv) The direction of magnetic flux lines inside the magnet is from the south pole to the north pole.

   Which of these statements are correct?

   a) i & ii b) i & iii c) iii & iv d) i & iv (1)

2. Statement : A pivoted magnetic needle placed near a current-carrying conductor experiences a deflection.

   Reason : A current-carrying conductor, when placed in a magnetic field, has a tendency to deflect.

   Which of the following options is correct?

   a) The statement and reason are correct; the reason explains the statement.

   b) The statement and reason are correct; however, the reason does not explain the statement.

   c) Both the statement and the reason are incorrect.

   d) The statement is incorrect, and the reason is correct. (1)

3. Two statements are given about the magnetic effect of electric current.

i) When an electric current flows through a conductor, a magnetic field is created around it.

ii) The direction of the magnetic field is determined by the direction of the electric current.

Choose the correct option:

a) i only is correct b) ii only is correct

c) i & ii are correct d) i & ii are wrong. (1)

4. Statement : The arrangement of magnetic flux lines around a current carrying solenoid is similar to that of a bar magnet.

Reason : The alignment of magnetic flux lines around a current carrying conductor does not depend on its shape.

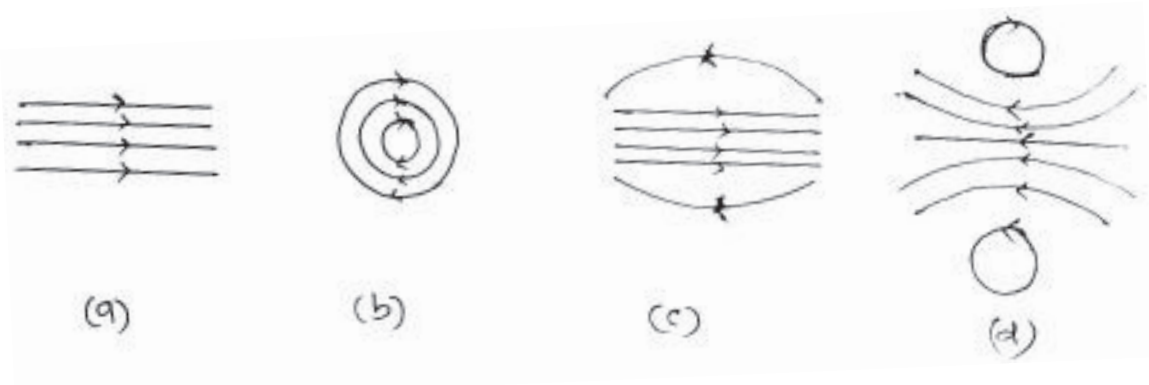
a) Statement and reason are correct; the reason explains the statement

b) Statement and reason are correct; however, the reason does not explain the statement

c) Both the statement and the reason are incorrect

d) Statement is incorrect reason is correct (1)

5. Which of the following is the arrangement of magnetic flux lines created by a current carrying solenoid? (1)



6. Choose the correct statement regarding the Magnetic polarity of a current carrying solenoid and write it down.

i) If the current in one end of the solenoid is clockwise, then that end is north pole.

ii) If the current in one end of the solenoid is clockwise, then that end is south pole.

iii) If the current in one end of the solenoid is anti clockwise direction, then that end is north pole.

iv) If the current in one end of the solenoid is anti clockwise direction, then that end is southpole.

a) a & c b) b & c c) a & d d) b & d

7. Observe the table and write the correct answer

A	B
i. Maxwell's right hand thumb rule	a. Find the direction of motion of a Current carrying conductor in a magnetic field.
ii. Fleming's left hand rule	b. Electric motor
iii. Motor principle	c. To find the direction of magnetic field around a current carrying conductor

a) i-b, ii-c, iii-a

b) i-c, ii-a, iii-b

c) i - a, ii-c, iii-b

d) i-a, ii-b, iii-c (1)

8. Name the device which is used to change the direction of current through the armature after each haf rotation. (1)

9. AB is a current carrying conductor. Based on the figure some statements are given.

i) The direction of magnetic field around the conductor AB is in anticlockwise direction.

ii) Maxwell's right hand thumb rule is used to find the direction of magnetic field.

iii) Direction of magnetic field in clockwise write the correct answer.

Write the correct answer

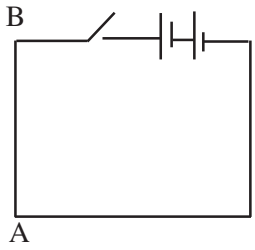


a) Only i correct b) i & ii correct

c) ii & iii correct d) i & iii correct (1)

10. Observe the picture of electricity flowing through the conductor AB.

Statement : Holding a conductor with your right hand in such a way that the thumb points in the direction of the electri current, the fingers curled around the conductor will indicate the direction of the magnetic field.



i) The magnetic field generated around the conductor AB is in the clockwise direction

ii) The direction of electric current through conductor AB is from A to B

iii) The direction of electric current through conductor AB is from B to A

iv) The magnetic field generated around the conductor AB is in the anti clock wise direction

a) i & ii are correct, statement is wrong

b) ii & iv are correct, statement is wrong

c) i & iii are correct, statement is correct

d) i & ii are correct, statement is wrong (1)

11. Statement 1 : Electromagnets are devices that create magnetic field using electricity.

Statement 2 : The magnetic strength of a current carrying solenoid increases with

decreases in the number of turns and decreases with increases in the amount of current.

Which one is correct among the following

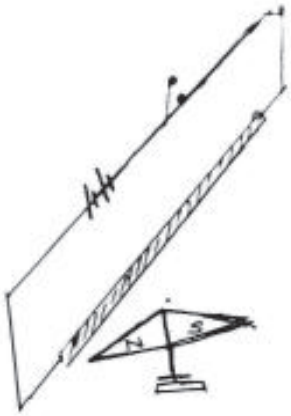
a) Statement 1 and 2 are correct

b) Statement 1 is correct but statement 2 is not correct

c) Statement 1 and 2 are not correct

d) Statement 1 is not correct but statement 2 is correct (1)

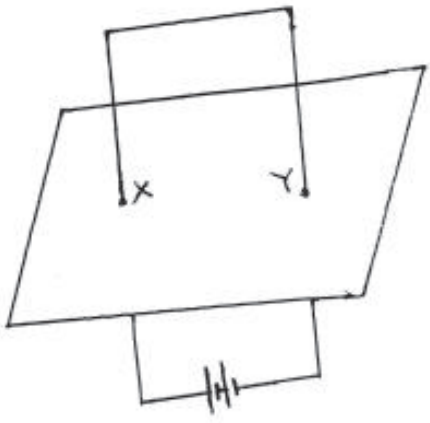
12. A conductor is placed vertically above a pivoted magnetic needle.



When the switch is turned on, in which direction will the north pole of the magnet needle be deflected? Give a reason. (2)

13. a) Draw the magnetic flux lines around the points X and Y in the figure and mark their direction. (1)

b) What is the change in the arrangement of magnetic flux lines if the intensity of the electric current through the conductor is increased? (1)



14. Choose the correct statement from the given ones.

i) The strength of the magnetic field of a coil of electrically conducting wire decreases as the number of turns increases.

ii) As the number of turns increases, the total flux around the wire increases.

iii) As the number of turns increases, the flux generated per turn increases.

iv) As the current through the coil of wire increases, the strength of the magnetic field increases.

a) i&ii b) ii&iii

c) iii&iv d) ii&iv (2)

15. The arrangement of magnetic field lines around a bar magnet is similar to the arrangement of magnetic field lines around a solenoid.

Therefore, the magnetic field created by a bar magnet and a solenoid is identical in everyway. Is this statement correct? justify . (2)

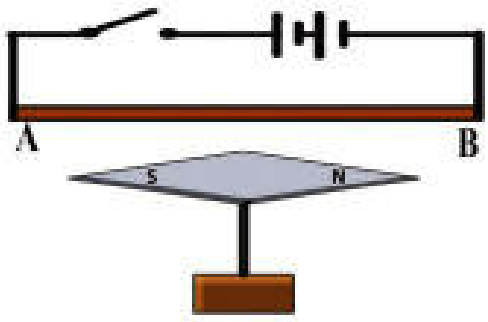
16. How is a bar magnet different from a current carrying solenoid? (2)

17. Write the name of two devices that work on the motor principle . (2)

18. Explain the working of moving coil loudspeaker speaker. (2)

19. If electricity flows in a conductor from east to west, find the direction of the magnetic field just above and below the conductor. (2)

20. Observe the figure.

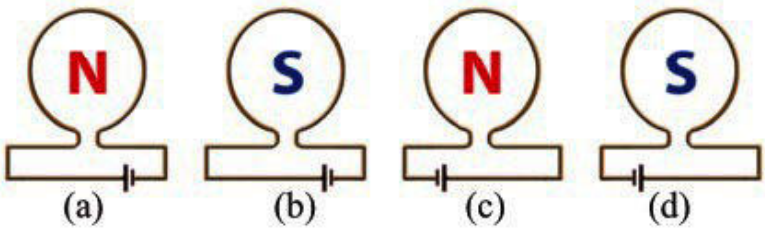


a) In which direction does the north pole of the magnetic needle deflect when the switch is turned on? (1)

(Clockwise direction / Anticlock wise direction)

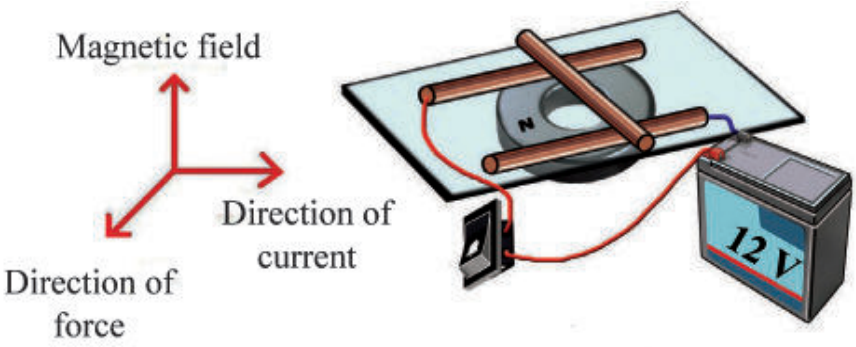
b) Suggest one method to change the direction of deflection of the magnetic neede. (1)

21. The figures of current carrying conducting loops are given below.



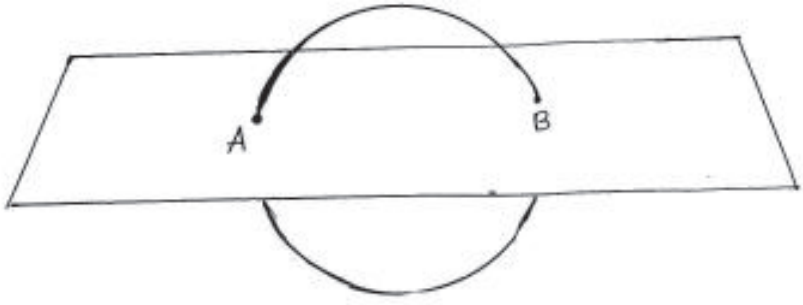
Which figures give the correct representation of the magnetic polarity of the end you are facing? Justify your answer. (2)

22. Based on the figure below. Write the factors that influence the direction of force experienced by the conductor? (2)



23. a) Connect a battery in the circuit shown below such that the current flows in an anti-clockwise direction on the side facing you. Draw the internal flux lines around points A and B and indicate their direction. (2)

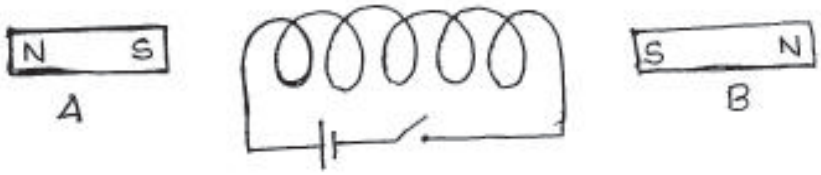
b) Suggest ways to increase the magnetic field of a current carrying conductor. (1)



24. a) How to determine the polarity of an electrically conductive solenoid

b) What are the factors that affect the magnetic field of a solenoid? (2)

25. The picture given shows two bar magnets placed near a current carrying solenoid.



a) When switched on

i) Magnet A attracts, magnet B repels.

ii) Magnet A repels, magnet B attracts.

iii) Magnets A and B come close to the solenoid.

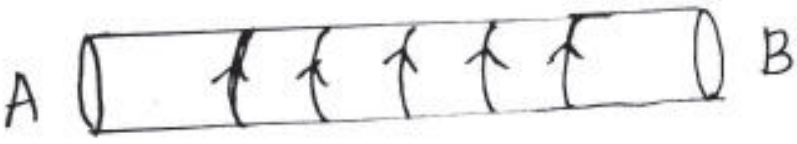
iv) Magnets A and B move away from the solenoid. (1)

b) Justify your answer. (2)

26. a) What is an electromagnet? (1)

b) Write any two uses of such magnets? (2)

27. The direction of magnetic field around a current carrying conductor AB is marked.

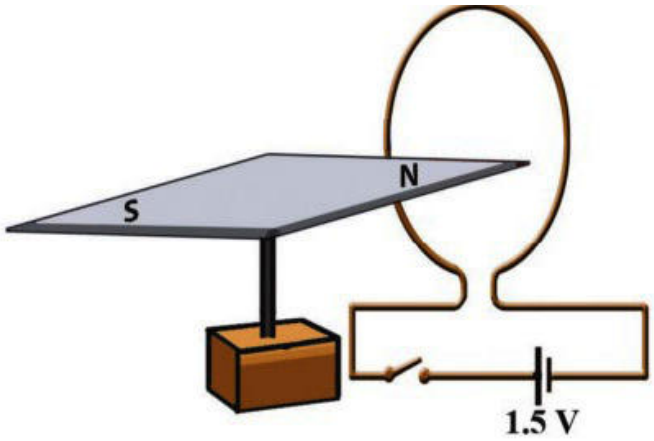


a) Find the direction of electric current through the conductor AB. (1)

(From A to B / From B to A)

b) State the law that support this. (2)

28. Figure shows a coil of wire which is connected to a battery and a belt switch.

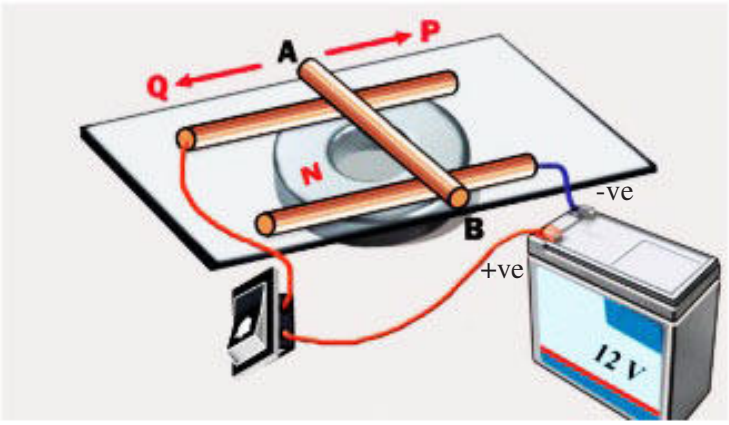


Hold the coil near one end of a pivoted magnetic needle.

a) What is your observation when the switch is turned on? What is the reason? (2)

b) Suggest two methods to increase the deflection of the magnetic needle (1)

29. Observe the figure and answer the following questions.



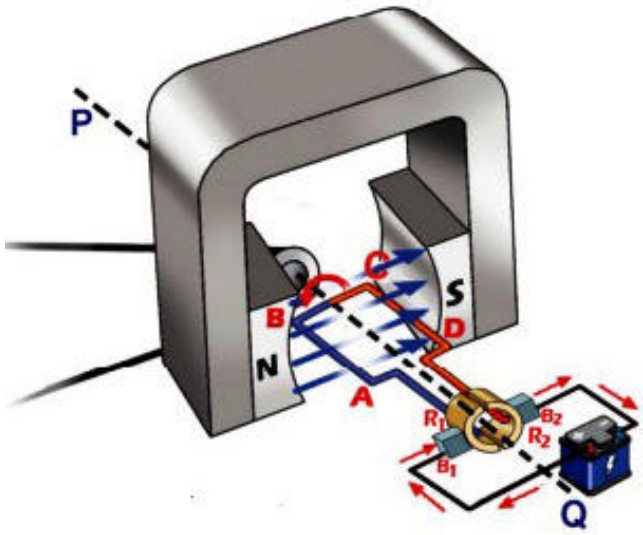
a) What is the direction of electric current through the conductor AB when the switch is turned on? (1)

b) In which direction does the conductor more. When current flows through the conductor AB? (1)

(Towards Q / towards P)

c) State the law that helps to find the direction of motion of the conductor? (1)

30. Electric motor is a device which connects electrical energy into mechanical energy based on motor principle.



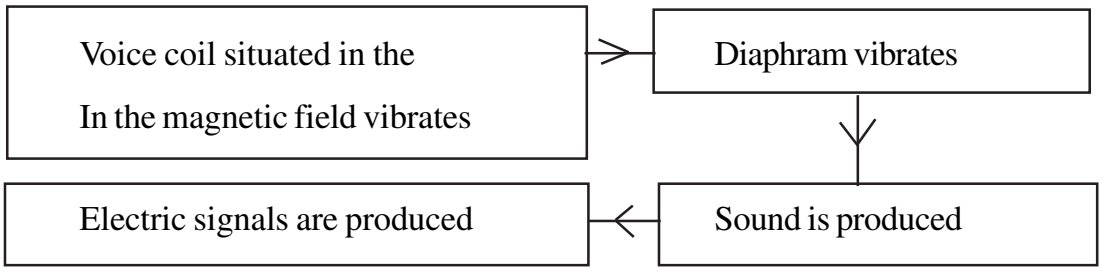
a) In which direction does the side AB and CD moves, when the direction of current through the armature coil ABCD is from A to D.

i - AB upward / CD downward (1)

ii - AB downward / CD upward

b) Justify your answer (2)

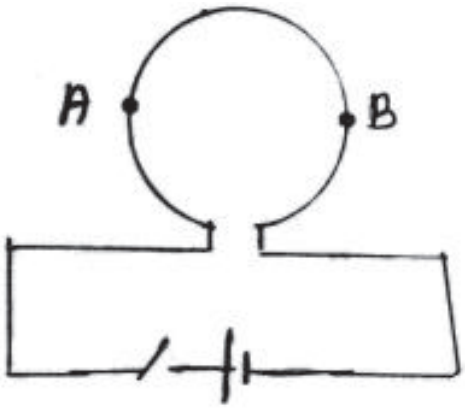
31. The working of a moving coil loudspeaker is given below in boxes.



a) Arrange them in proper sequence. (2)

b) Write the working principle of a moving coil loudspeaker. (1)

32. Observe the figure

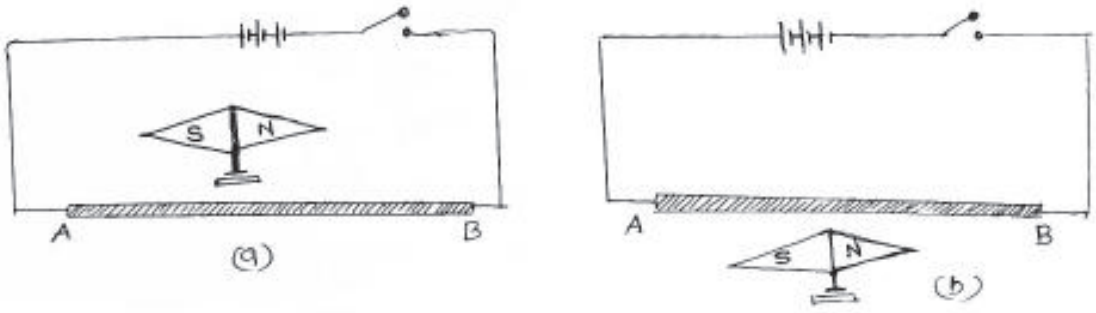


a) In which direction does the current flow in the coil that faces you, when the switch is turned on?

(Clockwise direction / Anti clockwise direction) (1)

b) In this case one student said that "the direction of the Hux lines is into the coil". Are you agree with this statement. Justify your answer. (2)

33. Observe the figures:



a) In each case, when the switch is turned on, is the north pole of the magnet placed next to the conductor AB deflected clockwise or anti-clockwise? (1)

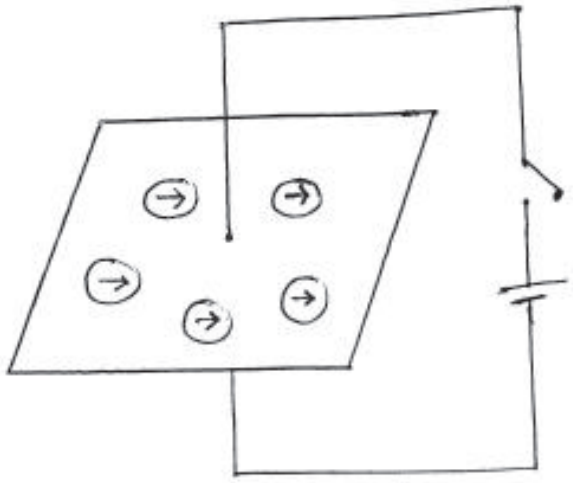
b) What is the reason for the deflection of the magnetic needle in each case? (1)

c) What is the change in the deflection of the magnetic needle in each of the following cases? (2)

i) Moving the magnetic needle away from the conductor

ii) The intensity of current through the conductor is reduced.s

34. Observe the figure

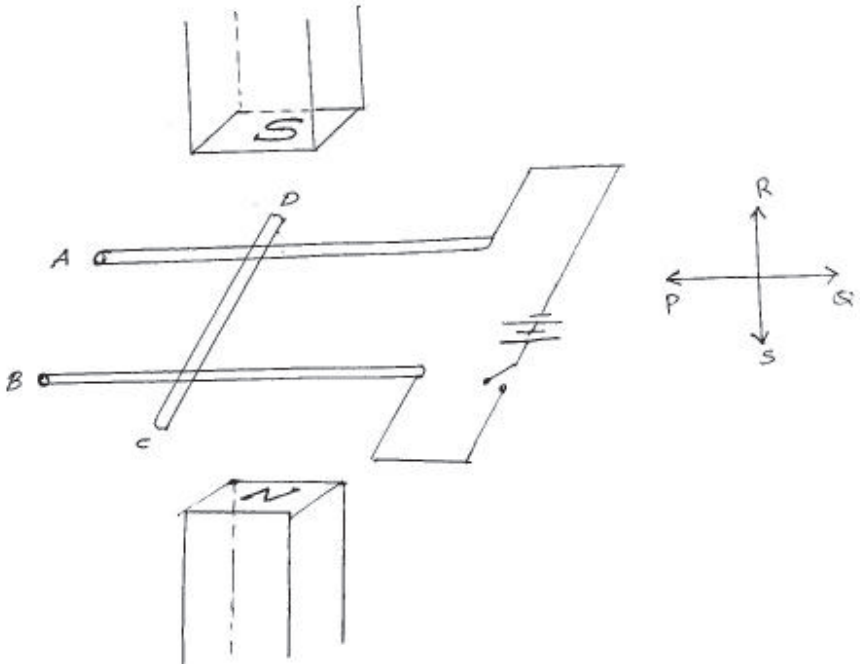


a) When the switch is turned on, what will be the direction of deflection of the north pole of the magnetic needle placed on the card board? (1)

b) Which rule helped you to determine the direction of rotation?state the rule. (2)

c) Draw the arrangement of the magnetic flux lines if the current through the conductor is reversed. (1)

35.



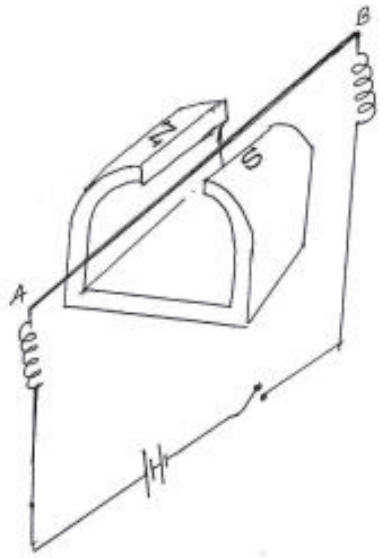
CD is a conductor rod arranged to be freely suspended above conductors AB.

a) What is the direction of movement of the CD when it is switched on? (1)

b) Which law helped you find the direction of motion? State the law. (2)

c) Suggest two ways to reverse the direction of motion of CD here. (1)

36.



A conductor AB is connected between two springs placed between strong magnetic poles.

a) When switched on:

i) The spring moves upwards.

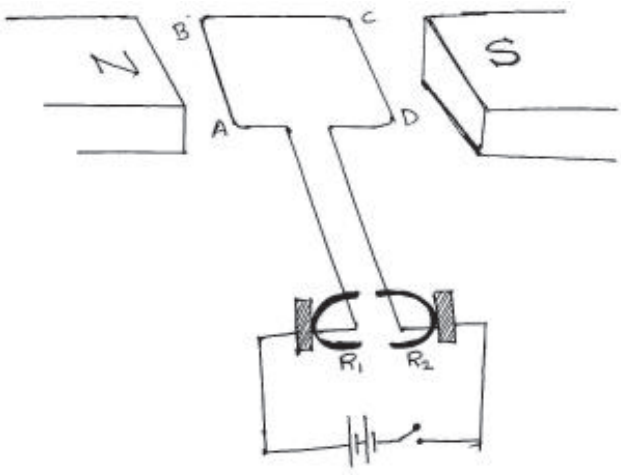
ii) The spring presses down

iii) Nothing changes (1)

b) Justify your answer. (1)

c) If the polarity of the magnet and the battery are reversed, what difference will be observed? Why? (2)

37.



a) What is the device shown in the picture? (1)

b) How will the movement of AB, CD sides be when the switch is turned on? (1)

c) What is the function of the part marked R_1, R_2 on this device? (1)

d) Write the name of another device that works on the same principle as this device. (1)

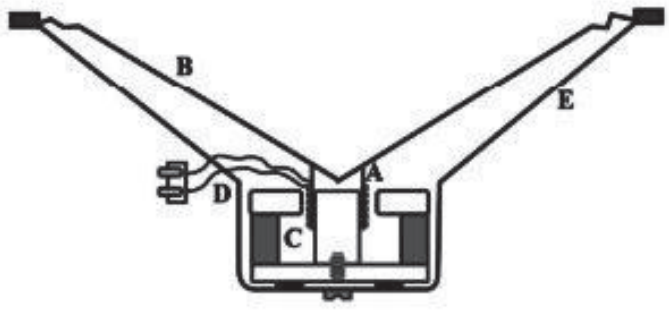
38. Wind 100 turns of insulated copper wire around an iron nail and connect the two end of the copper wire to a cell. Arrange some safety pins near the tip of the iron nail. It is seen that the safety pins are attracted by the iron nail.

a) Why the iron nail attract the safety pins? (1)

b) Write two methods to increase the magnetic strength of this device? (1)

c) Tabulate the difference between the magnetic field around a bar magnet and a current carrying solenoid? (2)

39. Observe the figure

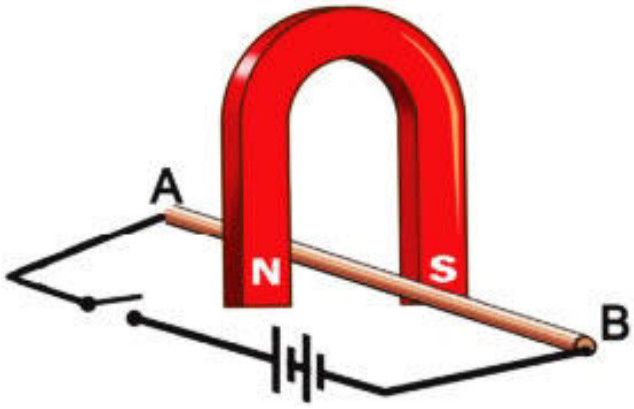


a) Identify the device shown in the figure and explain the working of this device (2)

b) What is the energy conversion taking place in this device? (1)

c) Name another device that works on the same principle. (1)

40.



In the figure, AB is conducting rod that is free to more.

a) When the bell switch is turned on, in which direction will the metal rod AB more? (1)

b) What should be done to keep the direction of motion of the rod unchanged while changing the direction of the magnetic field? (1)

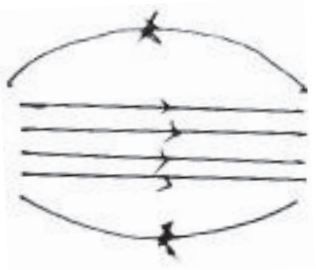
c) Which law used to find the direction of motion of the conductor in this case? (1)

Chapter 4

MAGNETIC EFFECT OF ELECTRIC CURRENT

ANSWERS

1. d) i & iv
2. b) The statement and reason are correct; however, the reason does not explain the statement.
3. b) ii only is correct
4. b) Statement and reason are correct; however, the reason does not explain the statement
5. c)



6. b) ii & iii
7. b) i-c, ii-a, iii-b
8. Split ring commutator
9. b) i & ii correct
10. c) i & iii are correct, statement is correct
11. b) Statement 1 is correct but statement 2 is not correct
12. The magnetic needle will not deflect because it is in the direction of the magnetic field line.
13. a) X-Anti clockwise direction
    Y-Clockwise direction
    b) Magnetic flux lines become more dense around the conductor.

14. d) ii&iv

15. No A bar magnet has permanent, fixed magnetic strength with unchangeable polarity, whereas a current carrying solenoid has temporary, adjustable magnetic strength with reversible polarity .

16. **Bar magnet**
Magnetism is permanent
Magnetic strength is fixed
Polarity cannot be changed

Solenoid
Magnetism is temporary
Magnetic strength can be varied
Polarity can be changed

17. DC Motor
Moving coil loudspeaker.

18. The electric signals (audio signals) received from a microphone are amplified using an amplifier. These audio signals are then passed through a voice coil, which is placed in a magnetic field. The coil experiences a force and vibrates because the coil carrying the electric current is placed in a magnetic field. This vibration causes the diaphragm to vibrate, thus reproducing the sound.

19. Above the conductor-towards the north
Below the conductor-towards the south

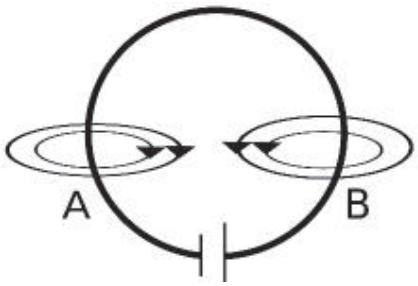
20. a) Anticlockwise
b) Reverse the direction of current
place the conductor above the magnetic needle.

21. b&c

If the current flows clockwise at one end of the solenoid, that end will be the south pole; if it flows anticlockwise, that end will be the north pole.

22. * Direction of current
    * Direction of magnetic field

23. a)



b) * increase current through the conductor
   * increase number of turns

24. a) If you hold a current carrying solenoid with your right hand in such a way that your four fingers curl the coils in the direction of the current, the thumb points towards the north pole of the solenoid.

b) * The number of turns of the conductor per unit length
   * Strength of current through the solenoid

25. a) ii) Magnet A repels, magnet B attracts.

b) The end facing A is the south pole of the solenoid and therefore repels the north pole of A. The end facing B is the north pole of the solenoid and therefore attracts the south pole of B.

26. a) Electromagnets are devices that create magnetic field using electricity.

b) * MRI Scanner
   * Cranes using electromagnet

27. a) B to A

b) Right hand thumb rule(state the rule)

28. a) Magnetic needle diflects. Conductor carrying electric current becomes an electromagnet, and a magnetic field is formed around it.

b) Increase the number of turns of the coil increase the current

29. a) A to B

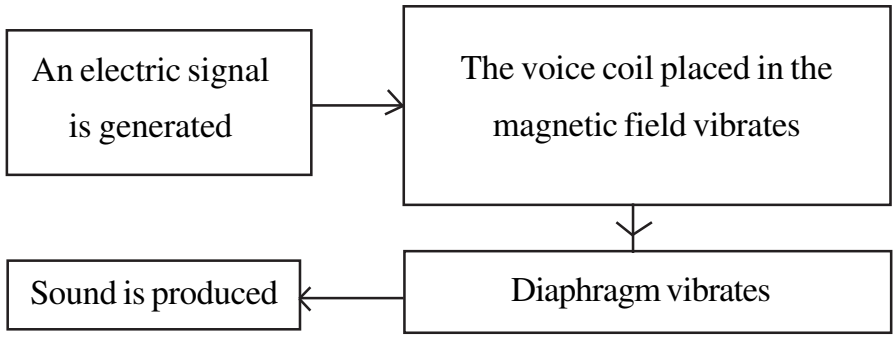
b) towards Q

c) Fleming's left hand rule.(state the rule)

30. a) ii - AB downward / CD upward

b) According to Fleming's left-hand rule, with current flowing from A to B in AB, the AB side experiences a downward force. With current flowing from C to D in CD, the CD side experiences an upward force, rotating the armature.

31. a)



b) Motor Laws

32. a) Clockwise direction

b) Agree. If the current in the coil is clockwise, the direction of the flux lines will be inward into the coil.

33. a) fig(a) - clockwise direction
fig(b) - anticlockwise direction

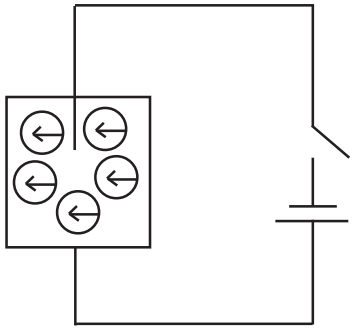
b) A magnetic field is formed around a current-carrying conductor. This magnetic field can exert a force on a magnetic needle, causing a deflection.

c) i) Deflection decreases ii) Deflection increases

34. a) Clockwise direction

b) Right hand thumb rule

c)



35. a) Towards Q

b) Fleming's left hand rule

c) Reverse the direction of current or magnetic field

36. a) ii) The spring presses down

b) Fleming's left hand rule

c) No difference. If the direction of the current and magnetic field are reversed together, the conductor will move in the same direction as before.

37. a) DC motor

b) AB - upwards

CD - downwards

c) Reverse the direction of current in armature ABCD ,ensuring continues rotation.

d) Moving coil loudspeaker.

38. a) Iron nail turns into an electromagnet.

b) * increase the number of turns of the coil

* increase the current

c) **Bar magnet**

Magnetism is permanent

Magnetic strength is fixed

Polarity cannot be changed

Solenoid

Magnetism is temporary

Magnetic strength can be varied

Polarity can be changed

39. a) Moving coil loudspeaker(explain the working)

    b) Electrical energy - mechanical energy

    c) DC motor

40. a) Moves upwards

    b) Reverse the direction of current.

    c) Fleming's left hand rule.(state the rule)

Chapter 5

Electric Energy : Consumption and Conservation

1 Mark

1. Various devices and the energy conversions that take place in them are given below. Choose the correct option

   i) Electric Kettle : Electric energy $\rightarrow$ Thermal energy.

   ii) Mixie : Electric energy $\rightarrow$ Mechanical energy.

   iii) Electric Iron : Electric energy $\rightarrow$ Light energy.

   iv) Microphone : Electric energy $\rightarrow$ Sound energy.

   a) All are correct. b) Only i, ii, iv are correct

   c) Only i, ii are correct d) Only i, iii, iv are correct

2. Nichrome is used as a heating coil in heating devices. Choose the correct one regarding the properties of nichrome.

   i) High oxidation resistance

   ii) Ability to provide heat energy for a long time in a red hot state

   iii) High resistivity

   iv) High melting point

   a) All are correct. b) Only i, ii, iv are correct

   c) Only i,ii, iii are correct d) Only iii, iv are correct

3. Which of the following is the equation for finding conductivity?

   ℓ/RA, RA/ℓ, $\ell/\rho A$, $\rho A/\ell$

4. Statement : The use of BLDC fans can reduce electricity consumption.

   Reason : Induction motor is not used in BLDC fans.

   a) The statement and the reason are correct. The reason explains the statement.

   b) The statement and the reason are correct. The reason does not explain the statement.

c) The statement and the reason are incorrect

d) The statement is correct. The reason is incorrect.

5. Statement : Energy crisis is the decrease in demand for energy and the increase in availability.

Reason : The use of fossil fuels is increasing as the population increases.

a) The statement and the reason are correct

b) The statement and the reason are incorrect

c) The Statement is correct but the reason is incorrect

d) The statement is incorrect but the reason is correct

6. Some physical quantities and their units are given. Choose the correct one

i) Resistance - Ω

ii) Resistivity - Ωm

iii) Electric energy - J

iv) Power - W

a) All are correct b) Only i, iii, iv are correct

c) Only i, ii, iv are correct. d) Only i, iv are correct.

7. Various equations are given to find electric power.

i) $P = I^2R$

ii) $P = V^2I$

iii) $P = \frac{V^2}{R}$

iv) $P = IR$

Choose the correct option

a) Only i, iii are correct b) Only ii, iv are correct

c) Only i,ii, are correct d) Only iii, iv are correct

8. Statement 1 : Electric heating appliances are appliances that utilize the heating effect of electricity.

Statement 2 : An electric kettle is a household appliance used for cooking.

a) Statement 1 and 2 are correct. Statement 2 is an example of statement 1

b) Statement 1 and 2 are correct. Statement 2 is not an example of statement 1

c) Statement 1 is correct but statement 2 is wrong

d) Statement 1 is false but statement 2 is true

9. Statement : Heating coil is a common part of electric heating appliances

Reason : Heating coil produces heat energy

a) The statement and the reason are correct. The reason explains the statement

b) The statement and the reason are correct. The reason does not explain the statement

c) The statement is correct. The reason is incorrect

d) The statement is incorrect. The reason is correct

10. Some of the factors that affect the resistance of a conductor are given below. Choose the correct option.

i) Length of conductor ii) Area of cross section

iii) Conductivity iv) Nature of material.

a) Only i, ii, iii are correct

b) Only i, ii, iv are correct

c) Only i, ii, are correct

d) All are correct

11. Statement : According to Joule's law $H = I^2Rt$.

Reason : If the voltage is constant, the quantity of heat produced increases when the resistance in the circuit is decreased.

a) The statement and the reason are correct, the reason explains the statement.

b) The statement and the reason are correct, the reason does not explain the statement.

c) The statement is correct, the reason is wrong.

d) The statement is wrong, the reason is correct.

12. Some statements are given regarding electric energy. Choose the correct one.

i) A watt-hour meter is a device used to measure electric energy used in homes.

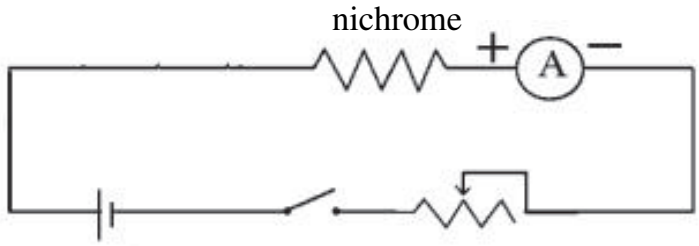
ii) The commercial unit of electrical energy is watt-hour.

iii) The SI unit of electric energy is J.

iv) One unit of electric energy is equal to 3.6×10^5J.

a) Only i, ii, iv are correct
b) Only i, iii, iv are correct
c) Only i, iii are correct
d) Only iii, iv are correct

13. Observe the given circuit. When the switch is turned on, the nichrome wire heats up. What are the ways to increase the heat generated in such circuits?



i) Use a longer nichrome wire

ii) Use a thicker nichrome wire

iii) Use an aluminium wire of the same size instead of nichrome

iv) Increase the current

a) ii, iii only
b) i, iv only
c) i, iii, iv only
d) ii, iii, iv only

14. Which of the following is the unit of conductivity?
$\left(\Omega m,\ \Omega/m,\ \Omega^{-1}m^{-1},\ \Omega m^{-1}\right)$

2 Marks

15. When the children said that increasing the resistance in a circuit will increase the amount of heat generated, only one child objected. He said that in some cases, increasing the resistance will reduce the heat. What is your opinion? Justify your answer.

16. Methane is a green house gas. Write another example of a green house gas. How do these affect global warming?

17. When solving a mathematical problem related to household appliances, it is preferable to use the equation $P = \frac{V^2}{R}$. Do you agree with this statement. Justify?

18. How much heat is produced if 0.2 A of current flows through a conductor with a resistance of 200Ω m for 5 minutes?

19. When 60 V is applied across the terminals of an electric heater, it passes a current of 4 A. If the potential difference is 90 V, what will be the current?

20. 'Energy is precious. Don't waste it.' What can you do to conserve energy?

21. If the resistance of a heating device operating at 230 Volt is 92Ω, how much heat is produced in 20 minute?

22. If an electric heater operating at 230 Volt produces 138000J of heat in 5 minute, what will be the current through the conductor?

23. Calculate the amount of heat generated when an electric heating appliance with a power of 1000W operates for 3 minute?

24. If the resistivity of a copper wire 5 m long and 1 millimeter in diameter is $\pi x10^{-8}\Omega$ m what will be the resistance of the conductor?

25. The resistance of a wire 2 meter long and 2 millimeter in diameter is 10Ω.
Find the resistivity.

3 Marks

26. The basis of all energy on earth is the sun. Solar panels are used to convert solar energy into electric energy.

    a) What are the benefits of installing solar panels?

    b) Solar production does not occur at night, so how can we use the energy from solar at night?

27. An electric bulb is marked as 220V, 100 W.

    a) What is its power when operated at 440V?

    b) Can this device be operated at 440 volts? Justify your answer.

28.

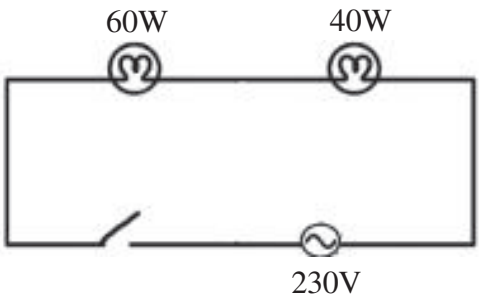


Figure 1

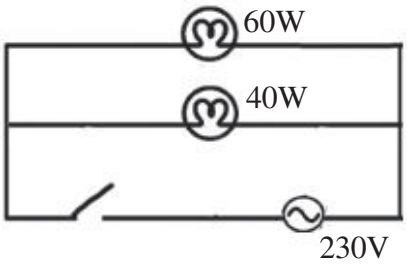


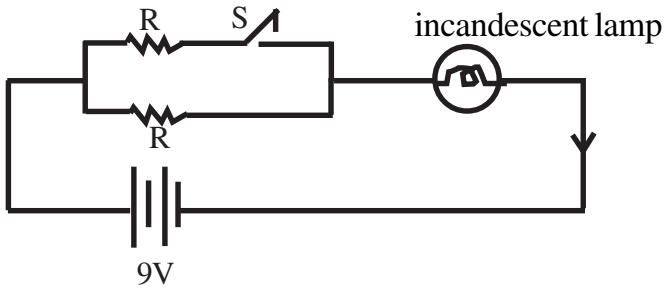
Figure 2

    a) If switched on, which bulb in Figure 1 is brighter?

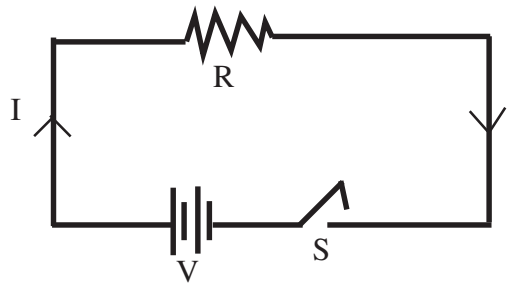
    b) If switched on, which bulb in Figure 2 is brighter?

    c) Which method in the figure should be used if you want to control each divices in your room using different switches?

29. An incandescent lamp glows due to the heat of the filament. What is the change in the brightness of the lamp when the switch is turned on and off in the circuit given below? Justify your answer?



30. Observe the circuit and find the answers.



200 J of heat is generated in this circuit per second.

a) What will be the change in current if the resistance (R) in the circuit is halved?

b) Then what will be the heat generated in the circuit?

c) State the law that helped you to find the answer

31. A heating coil with a resistance of 55Ω operates at 220 V.

a) Find the current in the circuit.

b) What will be the heat generated in 5 minute?

c) Find the power of the appliance?

32. 'To reduce atmospheric pollution, we should reduce our carbon footprint.' What is meant by carbon footprint? What can you do to reduce your carbon footprint?

4 Marks

33. In a house, a 1200 W Air conditioner operates for 5 hours, an electric iron of 500 W for an hour, 10 LED bulbs of 9 W for 10 hours and a grinder of 400 W for an hour.

a) What is the amount of electric energy consumed in a day?

b) If the appliances operate in the same manner, calculate the electricity charge for the month of September at a rate of Rs. 6.75 per unit.

34. Observe the circuits containing aluminium and nichrome wires.

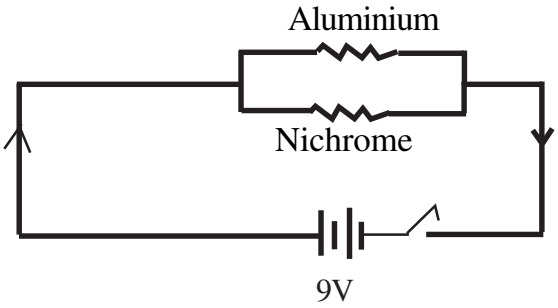


Figure 1

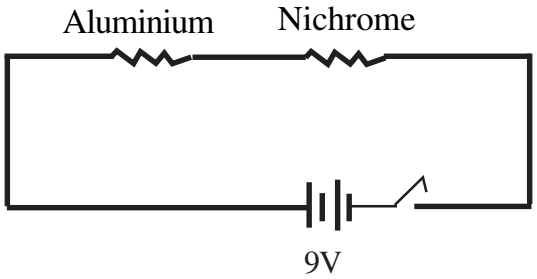


Figure 2

a) If switched on, which wire in the figure 1 generates more heat? Why?

b) Which wire in the figure 1 generates more heat? Why?

35. Electric heating appliances are devices that utilize the heating effect of electricity.

a) What is the name of the part used to produce heat energy in heating appliances?

b) Which metal alloy is it made of?

c) This metal alloy has high resistivity. How is this useful for heating appliances?

d) This metal alloy has very high oxidation resistance.What is oxidation resistance?

36. Power is the quantity of work done by an electrical appliance per unit time.

a) Which of the following is not a unit of power?

(Joule/second, Watt, Ohm metre, Volt ampere)

b) In our country, what will be the voltage at which electrical appliances operate to obtain the power labeled on electrical appliances?

c) Which equation of power is more preferable to use when solving numerical problems related to household applianced?

d) An electric heater produced 600000J of heat energy when operated for 10 minute. What would be the power of this heater?

37. Solar cells are devices that convert solar energy into electrical energy.

a) What is mean by solar panels?

b) Which is the first airport in the world to be operated entirely by solar energy?

c) What are the ways to incorporate the electricity generated from solar panels into the household circuit?

d) Which scheme has the government initiated to install solar panels in homes

38. Joule's law explains the factors that affect the quantity of heat produced in a conductor carrying an electric current.

a) What is the change in the heat produced if the current through the conductor is reduced by $\frac{1}{10}$?

b) What are the other two factors that affect the quantity of heat?

c) The resistance of a heating appliances operating at 230V is 115 Ω. What is the resistance required to double the heat energy produced by this device? Justify your answer?

•

Chapter 5

Electric Energy : Consumption and Conservation

Answers

1 Mark

1. c) Only i, ii are correct
2. a) All are correct
3. ℓ / RA
4. a) The statement and the reason are correct. The reason explains the statement
5. d) The statement is incorrect but the reason is correct
6. a) All are correct
7. a) Only i, iii are correct
8. a) Statement 1 and 2 are correct. Statement 2 is an example of statement 1
9. a) The statement and the reason are correct. The reason explains the statement
10. d) All are correct
11. b) The statement and the reason are correct, the reason does not explain the statement.
12. c) Only i, iii are correct
13. d) ii, iii, iv only
14. $\Omega^{-1}m^{-1}$

2 Marks

15. If voltage is constant,when the resistance in the circuit is increased, the current decreases.Hence the quantity of heat produced decreases.

16. Carbondioxide. Increase in green house gases cause global warming, resulting in climate change.

17. Agreed. Since the voltage is constant in home electrification, it is more preferable to use the equation $P = \frac{V^2}{R}$.

18. 2400 J

19. 6A

20. 1. Use energy efficient electrical appliances
    2. Turn off switches immediately after use
    3. Use LED tubes and bulbs
    4. Choose the size of the fan according to the size of the room
    5. Use BLDC fans..
    6. Promote public transport.(Any 4 points)

21. 690000 J

22. 2 A

23. 180000 J

24. $5x10^{-2}\,\Omega$

25. $20\pi x10^{-6}\,\Omega m$

3 Marks

26. a) 1. to reduce the energy crisis.
    2.to reduce environmental pollution.
    3.Solar panels are used for energy requirements in artificial satellites. (Any 2)
    b) The electricity obtained from such panels can be stored in a storage battery and utilised when needed.

27. a) 440W
    b) Not possible. If the voltage increases, the power increases and the bulb gets damaged.

28. a) 40W
    b) 60W
    c) Figure 2

29. When switched on, the resistors are in parallel and the effective resistance decreases. Then more electricity flows and the bulb lights up more intensely. When switched off, the resistance increases. Then the electric current intensity will decrease and the intensity of the bulb will decrease.

30. a) will double.

b) 400J

c) Joule's law.

31. a) 4A

b) 264000J

c) 880W

32. Greenhouse gases are directly and indirectly emitted by individuals, families, organizations, events, services, products, etc. The quantity of such emitted greenhouse gases, converted and expressed as equivalent to the measure of carbon dioxide, is carbon footprint.

1. Reduce domestic energy consumption.
2. Avoid wasting food.
3. Use public transport.
4. Reduce waste by utilising reusable products.

4 Marks

33. a) 9 unit

b) Rs. 1890

34. a) Aluminum. Since the wires are connected in parallel, more electricity flows through the low resistance aluminum and more heat is generated.

b) Nichrome. Since they are connected in series, the same electricity flows through the wires. Nichrome has higher resistance and therefore more heat is generated.

35. a) Heating element (Heating coil)

b) Nichrome

c) Even a nichrome wire of short length can provide sufficient resistance.

d) The ability of a material to resist corrosion due to contact with oxygen or other oxidisers at high temperature.

36. a) $\Omega\, m$

b) 230V

c) $P = \frac{V^2}{R}$

d) 1000 W

37. a) A solar panel is constructed by arranging many solar cells suitably to provide the electricity as per the requirement.

b) Cochin International Airport

c) 1. It can be stored in a storage battery and utilised when needed.

2. It can be directly supplied to the electricity distribution agencies.

d) Rooftop power project

38. a) Will be reduced to $\frac{1}{100}$

b) Resistancem time

c) $57.5\,\Omega$. Since the voltage is constant, heat increases as resistance decreases.

$H = \frac{V^2 t}{R}$

Chapter 6

ELECTROMAGNETIC INDUCTION IN DAILY LIFE

1 Mark

1. Observe the figure given below and find the correct answers from the following statements.



i) When the solenoid is moved towards the magnet, the magnetic flux change associated with the solenoid decreases.

ii) When the solenoid is moved towards the magnet, the magnetic flux variation associated with thesolenoid increases.

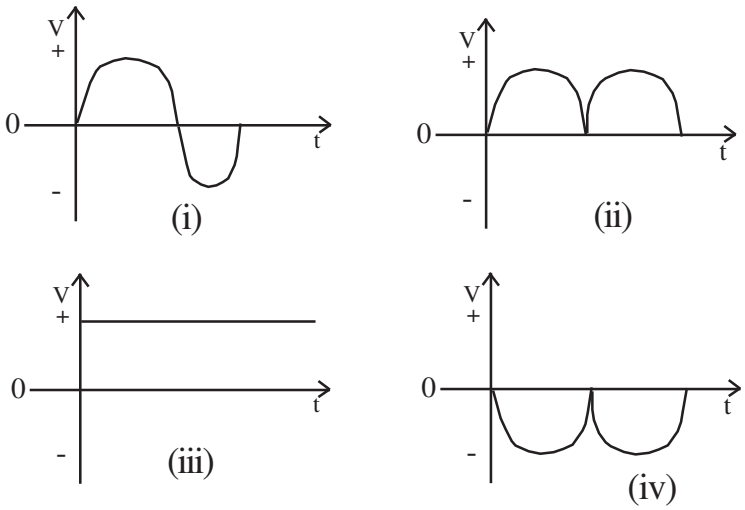
iii) When a magnet is placed on a solenoid and both are moved in the same direction at the same speed, the magnetic flux change associated with the solenoid is maximum.

iv) The magnetic flux change associated with the solenoid increases as the solenoid is pulled out from the magnet.

a) i &iv b) ii only

c) i&iv d) ii, iii & iv

2. Select the graphs below that are suitable for DC.



a) ii only b) ii & iii

c) iii only d) ii, iii & iv

3. Choose the most practical way to reduce the energy loss that occurs when electricity is transmitted over long distances through conductors.

i) Use the metal with comparatively low resistivity

ii) Reduce time

iii) Increase voltage and decrease current without changing power

iv) Use a suitable step-down transformer in power stations

a) i & iii b) ii & iv c) i & iv d) ii & iii

4. Select the correct equations relating the power of a transformer with no power loss.

i) $V_s \times I_p = V_p \times I_s$

ii) $\frac{I_s}{I_p} = \frac{V_p}{V_s}$

iii) $V_p \times I_p = V_s \times I_s$

iv) $\frac{V_s}{V_p} = \frac{I_s}{I_p}$

a) i & ii b) ii & iii

c) iii & iv d) iv & i

5. To what voltage is 11 kV reduced by a distribution transformer?

a) 110 V b) 230 V

b) 220 kV c) 115 V

6. Where is a step up transformer used in the following?

a) Power station b) Substation

c) Secondary substation d) Distribution transformer

7. Select the correct statements related to transformers at different stages of power distribution.

i) A step-down transformer is used in a power station.

ii) The transformer at the substation reduces 220 kV to 110 kV.

iii) 220 kV electricity reaches homes through a distribution transformer.

iv) Step-down transformers are used in substations.

a) i & ii b) ii & iii

c) ii & iv d) iii & iv

8. Statement : The electricity generated at 11kV at the power station is being increased to 220 kV.

Reason : By increasing the voltage without changing the power, we can over come the energy loss encountered when transmitting electricity over long distances through conductive wire

a) Statement and reason are correct; the reason explains the statement

b) Statement and reason are correct; however, the reason does not explain the statement

c) Both the statement and the reason are incorrect

d) Statement is incorrect reason is correct

9. Choose the correct statement about household electrification.

i) The main fuse is connected in parallel between the watt hour meter and the main switch.

ii) The earth line starts from the main switch.

iii) An appliance and the switch that control it is connected in series to the phase line.

iv) For high power appliances separate branch wires should be used with thick wires.

a) i & ii b) ii & iii

c) iii & iv d) iv & i

10. Statement : The voltage on the secondary of a step-up transformer will be lower and the current higher than that on the primary.

Reason : In a transformer with no power loss the power in the primary is equal to the power in the secondary.

a) Statement and reason are correct; the reason explains the statement

b) Statement and reason are correct; however, the reason does not explain the statement

c) Both the statement and the reason are incorrect

d) Statement is incorrect reason is correct

11. Select thermal power stations from the following

i) Thehri ii) Kalpakkam iii) Mettur iv) Brahmapuram

a) i & ii b) ii & iii

c) iii & iv d) i & iv

12. Statement : When the solenoid connected to the galvanometer is moved rapidly in and out of the magnet, the galvanometer needle deflects.

Reason : When the flux in a closed circuit changes, an emf is induced in the circuit.

a) Statement and reason are correct; the reason explains the statement

b) Statement and reason are correct; however, the reason does not explain the statement

c) Both the statement and the reason are incorrect

d) Statement is incorrect reason is correct

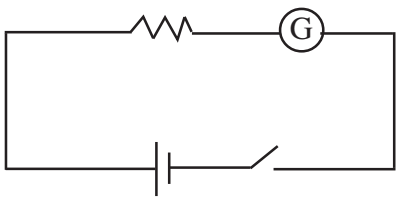
2 Marks

13. Whenever the magnetic flux linked with a closed circuit changes, an emf is induced in the circuit.

a) What is this phenomenon? (1)

b) Write two factors that influence the magnitude of the induced emf. (1)

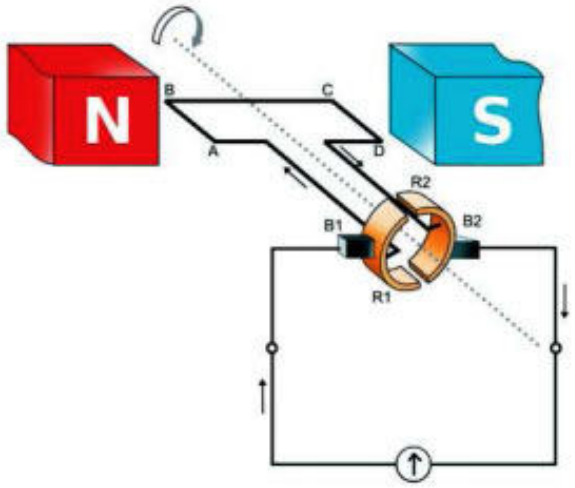
14. Observe the circuit given below.



a) What type of electricity is generated in the circuit? (1)

b) Write two devices that work using this electricity. (1)

15. A diagram of a DC generator is given.



Draw a diagram showing the change in its structure during the next half - rotation. (2)

16. Centres which produce electricity on a large scale for distribution are power stations.
    a) What fuels are used in thermal power stations? (1)
    b) What is the energy change that takes place in a nuclear power station? (1)

17. Hydroelectric power stations are power stations that generate electricity from water.
    a) Write the energy transformations that take place in a hydroelectric power station in order. (1)
    b) What are the two important hydroelectric power stations in India? (1)

18. Write down the methods to minimise energy loss while transmitting electricity over long distance through conducting wires. (2)

19. Transformer change AC voltage without change in power.
    a) What is the working principle of a transformer? (1)
    b) In which type of transformer will the secondary current lower than the primary current? (1)

20. Explain the reason for increasing the voltage of electricity generated at 11 kV at the power station to 220 kV based on the Joule's law. (2)

21. Draw a circuit diagram for constructing a branch circuit with MCB, two bulbs, one two pin socket and necessary switches. (2)

22. A safety fuse is a device that works based on the heating effect of electricity.
    a) Which are the situations that could lead to excessive current causing the fuse wire to melt? (1)
    b) What are the characteristics that a fuse wire should have? (1)

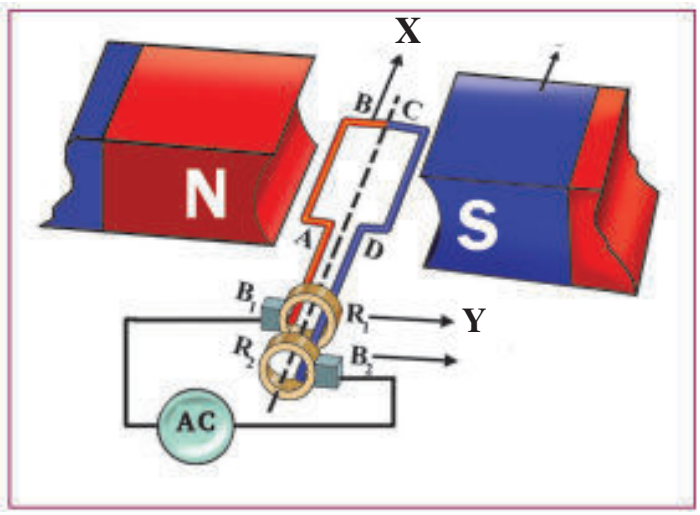
23. Explain how the three-pin plug ensures safety in household electrification. (2)

24. Electric shock is the impact caused by the flow of current through the body. What is the first aid to be given in case of electric shock? (2)

25. MCB and ELCB are two methods of ensuring safety in household electric circuit, but explain how they differ in their functions. (2)

3 Marks

26. The schematic diagram of an AC generator is given.



    a) What is the working principle of a generator? (1)
    b) Find X and Y. (1)
    c) What changes should be made in the structure to convert the AC induced in the mature in to DC for the external circuit.

27. Centres which produce electricity on a large scale for distribution are power station.

a) 75% of the electricity generated in India is generated with the help of which type of power station? (1)

b) How does a nuclear power station differ from a thermal power station? (2)

28. Thermal power accounts for about 75% of the electricity generated in India.

a) To reduce environmental pollution, electricity generation from natural sources such as Solar energy and tidal energy needs to be increased. Do you agree with this statement? Justify? (1)

b) Write down the diadvantages of each of solar energy and tidal energy? (2)

29. AC generated at power stations reaches our homes through transmission lines.

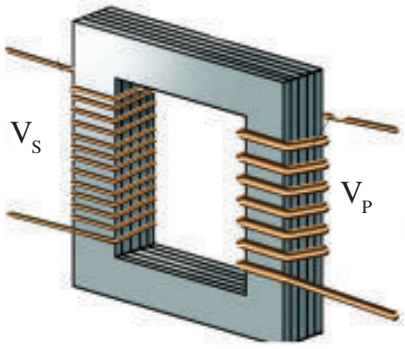
a) How does energy loss occur in this case? (1)

b) Suggest solutions to minimize this energy loss. (2)

30. Transformer is a device that works on the principle of mutual induction.

a) Identify the given transformer. (1)

b) What are the structural difference between step up transformer and step down transformer?. (2)

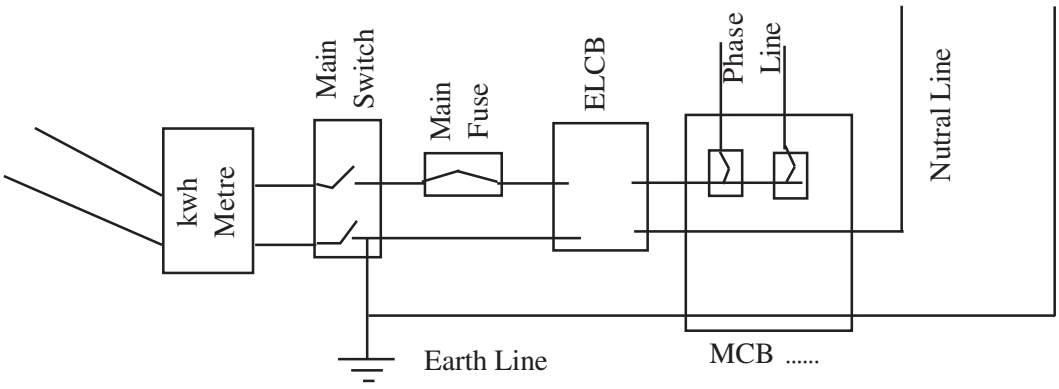


31. A transformer with no power loss has 2000 turns in the primary and 200 turns in the secondary. If 220V and 2A current are supplied to the primary

a) Calculate the power in seconds. (1)

b) Calculate the secondary voltage. (2)

32. Observe the diagram of the household wiring.



a) If there are any errors in the given circuit, find them and correct them. (2)

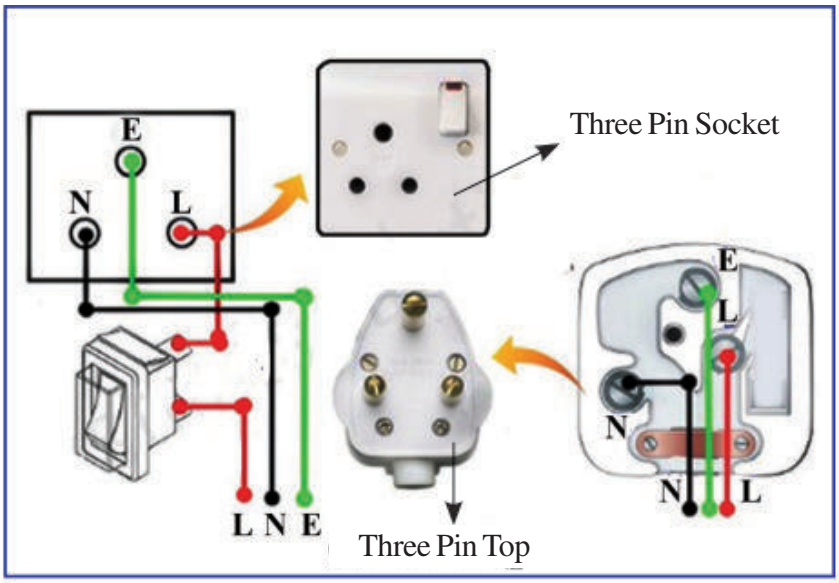
b) What is the role of a fuse in household wiring? (1)

33. A transformer has 200 turns in its primary coil and 800 turns in its secondary coil. If the input voltage is 220V, then

a) Calculate the output voltage. (2)

b) Identify the type of transformer (1)

34. A diagram of a three pin plug is given.

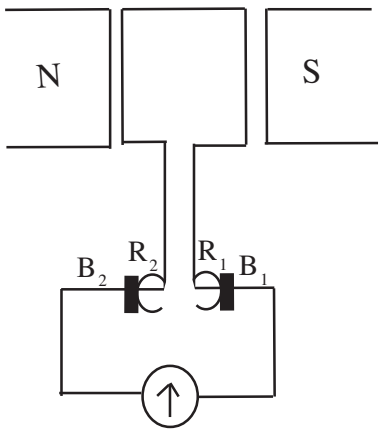


a) Where should the earth line marked with the letter E be attached to the appliance? (1)

b) Explain how the three pin ensures safety. (2)

4 Marks

35. A diagram of a DC generator is given below



a) Write the energy change takes place in the generator. (1)

b) Draw a graphical representation of the electricity produced in the given generator. (1)

c) What type of electricity will be produced in the circuit if the armature of this generator is stationary and the field magnet is rotated? (2)

36. There are many measures to ensure safety in domestic electricity distribution.

a) What is the function of the main fuse in a household electric circuit? (1)

b) Which device is used in a branch circuit instead of a fuse? Explain its function. (2)

c) Which device helps in automatically disconnecting the circuit if there is a current leak in the circuit? (1)

37. Electric shock is the impact caused by the flow of current through the body.

a) What are the precautions to be taken to avoid electric shock. (2)

b) What is the first aid for electric shock? (2)

38. The primary and secondary currents of a transformer are 2A and 6A respectively. Its output voltage is 120 volts. Then,

a) Which type of transformer is this. (1)

b) What will be the power in the primary? (1)

c) Find the number of turns in the secondary if the primary has 300 turns. (2)

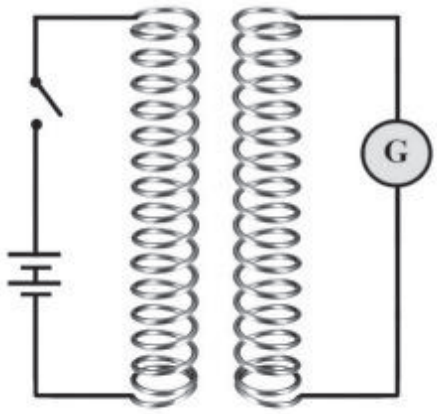
39. The diagram of a circuit made using a 9V DC source, switch, galvanometer, and coils is given.

a) Write any two ways to make the galvanometer needle diflect. (1)

b) What is this phenomenon called? (1)

c) What changes are needed for continuous deflection? (1)

d) Name a device that works on this principle. (1)



40. When electricity is transported from a power station to distant locations through transmission lines, energy is lost as heat.

a) How can this energy loss be minimized? (1)

b) Which device helps in varying voltage without loss of power? (1)

c) Explain the working principle of this device. (2)

41. In a physics class, a child suggested that it is possible to generate AC electricity without changing the structure of a DC generator.

a) Do you agree with this statement? Justify your answer. (2)

b) What is the function of split rings in a DC generator? (1)

c) What are the ways to get the energy needed to work the generator in a power station? (1)

Chapter 6

ELECTROMAGNETIC INDUCTION IN DAILY LIFE

ANSWERS

1 Mark

1. c) ii & iv
2. d) ii, iii & iv
3. a) i & iii
4. b) ii & iii
5. b) 230 V
6. a) Power station
7. c) ii & iv
8. a) Statement and reason are correct; the reason explains the statement
9. c) iii & iv
10. d) Statement is incorrect reason is correct
11. c) iii & iv
12. a) Statement and reason are correct; the reason explains the statement

2 Mark

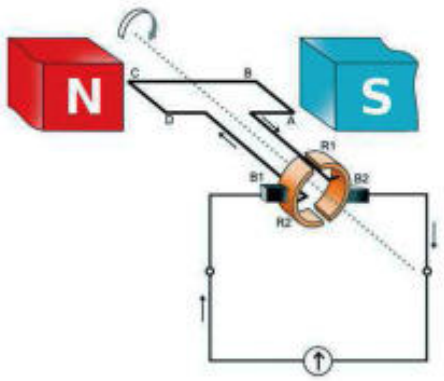
13. a) Electro magnetic induction (1)

    b) * Number of turns per unit length
    * Strength of magnet
    * Speed of relative motion (1)

14. a) DC (1)

    b) Mobile phone (1)

15.



16. a) CNG, Coal (1)

b) Nuclear energy → Electrical energy (1)

17. a) Potential energy → Kinetic energy → Mechanical energy → Electrical energy (1)

b) Moolamattom, Pallivasal, Thehri (1)

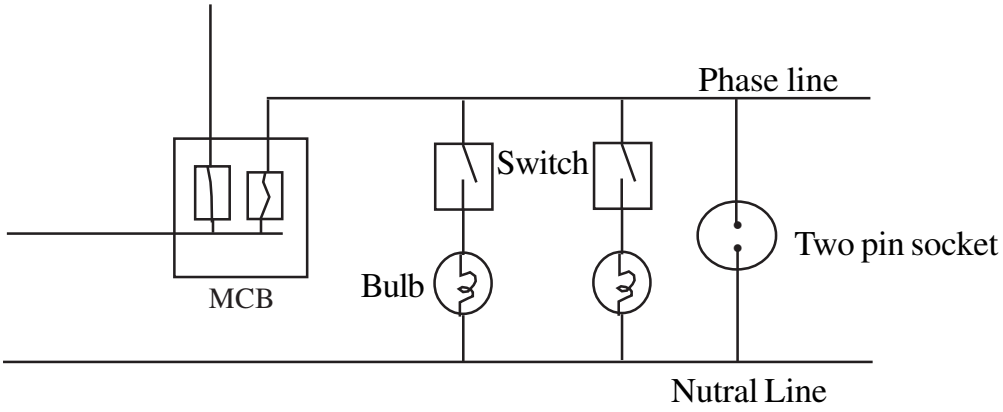
18. • Use suitable metal wire with low resistivity
• Increase voltage and decrease current without changing power (2)

19. a) Mutual induction (1)

b) Stepup transformer (1)

20. The voltage of electricity generated at 11 KV is increased to 220 KV for transmission because, according to Joule's law, energy loss in transmission lines is proportional to the square of the current ($H = I^2Rt$). By increasing the voltage, the current decreases for the same power transmission ($P = VI$), reducing energy loss as heat in the lines. (2)

21.



(2)

22. a) Over loading,short circuit. (1)

b) • low melting point (1)
• An alloy of tin and lead

23 .The three-pin plug has three wires: live, neutral, and earth. The earth pin connects the metal body of an appliance to the ground. If there's a fault and the live wire touches the

metal body, the earth wire provides a safe path for the current to flow to the ground, tripping the circuit breaker or blowing a fuse, preventing electric shock. (2)

24. • Rub the body to increase blood circulation and raise body temperature.
    • Administer artificial respiration.
    • Rub the muscles to restore them to their normal state (2)

25. When there is excess current in circuit due to short circuit or overload MCB automatically operates and disconnect the circuit.ELCB trips the circuit when it detects a leakage current to earth , protecting against electric shock. (2)

3 Mark

26. a) Electro magnetic induction (1)
    b) X-Armature Y-Slip ring (1)
    c) Use split ring instead of slip ring (1)

27 a) Thermal power station. (1)
Thermal power station is a power station that generates electricity by operating a generator using fuels like CNG, coal, etc.
Nuclear Power Stations generate electricity by using nuclear energy to convert water into steam. This rotates turbines, which in turn operate generators to generate electricity (2)

28. a) Agreed. Increasing electricity generation from natural sources can reduce green house gas emission and reduce pollution. (1)
    b) * Solar energy is only practical in places and situations where there is good sunlight.
    * Generating electricity from tidal energy is only practical if the tides in coastal area exceed a certain height. (2)

29. a) Energy loss in the form of heat. (1)
    b) • Use suitable metal wire with low resistivity
    • Increase voltage and decrease current without changing power (2)

30. a) Step up (1)

b)

Stepup transformer	Stepdown transformer
Thick wires are used in the primary	Thin wires are used in the primary
The primary has lesser number of turns than the secondary	The secondary has lesser number of turns than the primary

(2)

31. a) In a transformer with no power loss , the power in the primary is equal to the power in the secondary

Secondary power = 240 x 2 = 480W (1)

b) $N_p = 2000$ $N_s = 200$ $V_p = 240V$

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

$$V_s = \frac{N_s x V_p}{N_p} = \frac{200x240}{2000} = 24 \text{ V}$$ (2)

32. a) * Main fuse should be connected between the watt hour meter and the main switch.

* Earth line starts from Kwh meter (2)

b) Excessive current flowing through a circuit due to over loading, short circuit lightning etc, mets the fuse wire and breaks the electrical connection. (1)

33. a) $N_p = 200$, $N_s = 800$, $V_p = 220V$, $V_s = ?$

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

$$V_s = \frac{N_s}{N_p} xV_p$$

$$= \frac{800}{200} \text{x} 220$$

$= 4x200 = 880V$ (2)

b) Since Ns > Np and Vs > Vp, it is a stepup transformer. (1)

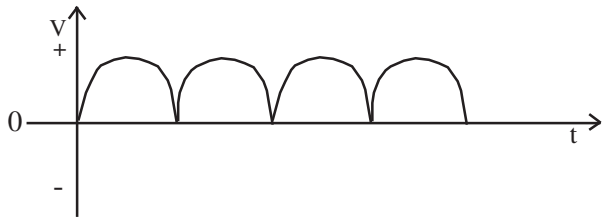
34. a) The earth line should be attached to the metal body of the appliance. (1)

b) The three-pin plug has live, neutral, and earth pins. The earth pin connects an appliance's metal body to the ground. If a fault occurs and the live wire touches the metal body, the earth wire provides a safe path for current to flow to the ground, tripping the circuit breaker or blowing a fuse, preventing electric shock (2)

4 marks

35. a) Mechanical energy $\rightarrow$ Electrical energy (1)

b)



(1)

c) AC. Since the positions of the split rings do not change each other, the same AC induced in the armature is obtained in the external circuit. (1)

d) AC electricity will be produced. The rotating magnet still creates a changing magnetic field in the stationary coils, inducing an alternating current. (1)

36. a) Overloading, short circuits, lightning, etc., can cause excessive current flow through a circuit. A safety fuse is a device to protect living beings and equipment from the dangers caused by this. (1)

b) MCB is a device used in branch circuits instead of safety fuse. When there is excessive current in a circuit due to short circuit or overload, the MCB auto matically operates and disconnects the circuit ie., it trips. After resolving the circuit problem, the MCB switch can be turned on to restore the circuit to its original state. (2)

c) ELCB (1)

37. a) * Do not handle electrical appliances or operate switches with wet hands.
* Plug into or unplug from a socket only after turning off the switch. (2)

b) * Administer artificial respiration.

* Rub the muscles to restore them to their normal state. (2)

38. a) Stepdown transformer (1)

b) Power = 120 x 6 = 720 W (1)

c) Vp x Ip = Vs x Is

$$V_p = \frac{V_s x I_s}{I_p} = \frac{120x6}{2}$$

= 360V

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

$$N_s = \frac{V_s x N_p}{V_p} = \frac{120x300}{360} = 100 \text{ turns}$$ (2)

39. a) While the switch turned on and off (1)

b) Mutual induction (1)

c) Supply AC instead of DC in first coil (1)

d) Transformer (explain) (1)

40. a) * Use suitable metal wire with low resistivity

* Increase voltage and decrease current without changing power (1)

b) Transformer (1)

c) Mutual induction (2)

41. a) Yes. Agree. if the armature of DC generator is stationary and the field magnet is rotated ,AC will be obtained in the output. (2)

b) Split rings in a DC generator convert AC to DC by reversing the current direction every half cycle. (1)

c) Thermal energy

Nuclear energy

hydro electric energy etc.. (1)

Chapter 7

MECHANICAL ADVANTAGE IN ACTION

1 Mark

1. Some statements are given.
   i) Changes the magnitude of the effect of the force.
   ii) Changes the direction of the applied force.
   iii) Changes the magnitude of the effect of force and the direction of the force.
   iv) Changes the magnitude of the effect of force without changing the direction of the force.

   Choose the correct one for simple machines.
   a) i, ii, iii only b) i, ii only
   c) iv only d) All

2. Statement : In a second order lever, the load is always greater than the effort.
   Reason : In a second order lever, the effort arm is always greater than the load arm.

   Which of the following options is correct?
   a) The statement and the reason are correct, the reason explains the statement
   b) The statement and the reason are correct, the reason does not explain the statement
   c) The statement is correct and the reason is wrong
   d) The statement is wrong and the reason is correct

3. Which one is correct regarding the first order lever?
   i) Fulcrum is in between the load and effort.
   ii) Load is in between the effort and the fulcrum
   iii) Effort is in between the load and effort.
   iv) Effort and Fulcrum will be close together
   a) i only b) ii only
   c) iii only d) i and iv only

4. lf effort arm and load arm are equal, then Mechanical Advantage is,
   a) Zero b) Between zero and one
   c) One d) More than one

5. Select the one with Mechanical Advantage is always greater than one.

i) First order lever

ii) Second order lever

iii) Third order lever.

iv) Moving pulley

a) i, ii, iv only b) i, ii, iii only

c) ii,iv only d) i, iv only

6. When lifting objects using a fixed pulley,

a) Can reduce the magnitude of the applied force

b) Can change the direction of the applied force

c) Can increase the magnitude of the load

d) Can reduce the magnitude of the applied force and change the direction of the applied force.

7. Statement : A movable pulley is a second order lever.

Reason : In a movable pulley, the effort arm is shorter than the load arm.

Which of the following options is correct?

a) The statement and the reason are correct, the reason explains the statement.

b) The statement and the reason are correct, the reason does not explain the statement.

c) The statement is correct and the reason is wrong

d) The statement is wrong and the reason is correct

8. A gear is a mechanical device used in vehicles.

Some statements related to gear are given.

i) To increase the speed of the vehicle, the toothed wheel attached to the engine must be connected to the small toothed wheel that helps to turn the tyre of the vehicle.

ii) While going uphill, the toothed wheel attached to the engine should be con nected to a large toothed wheel.

iii) When the toothed wheel attached to the engine connected to a large toothed wheel, its efficiency to rotate decreases.

iv) When connected to a large toothed wheel, the speed of the vehicle decreases.

Choose the correct one from the following options?

a) i, ii, iv only b) i, ii, iii only

c) iii, iv only d) All

9. To increase the mechanical advantage of an inclined plane,

i) The length of the inclined plane should be increased

ii) The height of the inclined plane should be decreased

iiii) There should be less effort and more load

iv) There should be less load and more effort

a) i, ii, iv only b) i, ii, iii only

c) i, iv only d) ii, iii only

10. Statement : The mechanical advantage of a screw is $= \frac{\text{Length of one thread}(\ell)}{\text{Pitch(h)}}$

Reason : The screw is considered as an inclined plane.

Which of the following options is correct?

a) The statement and the reason are correct, the reason explains the statement

b) The statement and the reason are correct, the reason does not explain the statement

c) The statement is correct and the reason is wrong

d) The statement is wrong and the reason is correct

11. Write the name of two simple machines whose mechanical advantage is greater than 1.

12. Statement 1 : In first class levers, the fulcrum is always between the effort and the load

Statement 2 : In second class levers, the mechanical advantage is always less than one.

Which of the following options is correct?

a) Statements 1 and 2 are correct

b) Statement 1 is correct 2 is incorrect

c) Statement 1 is incorrect 2 is correct

d) Statements 1 and 2 are incorrect

13. Two children of different weights are playing on a seesaw. The heavier child uses load and the lighter child uses effort. If the children want to continue playing easily,

a) The heavier child should sit closer to the fulcrum.

b) The heavier child should sit away from the fulcrum.

c) The lighter child should sit closer to the fulcrum.

d) Both should be at the same distance.

14. Statement 1 : The mechanical advantage of a fixed pulley will always be 1

Statement 2 : The effort of a moving pulley will be half of the load(1)

a) Statement 1 and 2 are correct

b) Statement 1 is correct 2 is incorrect

c) Statement 1 is incorrect 2 is correct

d) Statement 1 and 2 are incorrect

2 Marks

15. A person uses a wheel and axle to lift a bucket from a well. If the diameter of the wheel is 50 cm and the radius of the axle is 5 cm, what is the mechanical advantage in this situation?

16. A gear is a mechanical device used in vehicles. How does the gear work while going uphill and to increase the speed of the vehicle?

17. Gears are used in vehicle to transfer motion or force from one part of the machine to another.

a) A car going uphill, the toothed wheel attached to the engine should be connected to which wheel of the axle?

i) small wheel

ii) medium wheel

iii) large wheel

iv) any wheel

b) Justify your answer

18. A truck that had fallen into a lake was brought to shore using a wheel and axle. One person commented that the task would have been easier if they had used a wheel with a larger radius. Do you agree with this? Why?

19. Classify the following statements suitably for fixed pulley and movable pulley.
    i) Can change the direction of the applied force only
    ii) Can reduce the magnitude of the applied force
    iii) Mechanical advantage is one
    iv) Mechanical advantage is 2

20. "No gain in work is achieved while using a moving pulley".
    Do you agree with this statement? Justify your answer.

21. A child needs to lift an object of 1.5 m high. Is a 2 m or 4 m inclined plane suitable? Explain.

22. A crowbar is a simple machine used to lift heavy objects by applying a small force at one end.
    a) Which order of lever is a crowbar?
    b) If a crowbar of length 1.5 m is used to lift a stone, and the fulcrum is 0.5 m away from the stone, calculate the mechanical advantage of the crowbar.

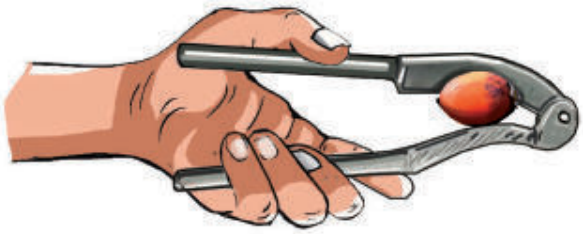
23. Simple machines are devices that make the extension easier. What is the mechanical advantage of a simple machine? What are two simple machines used in daily life?

24. Workers use a 6m long inclined plane to load large piece of wood onto a 1m high lorry. What is the mechanical advantage from this?

25. What is the effort required to lift a load of 60N using a wheel of radius 18 cm and an axle of radius 4.5 cm?

3 Marks

26.



a) Which order of lever does the figure represent?

b) What is its mechanical advantage?
(Zero, between zero and one, one, greater than one)

c) Write another example for the same order lever?

27. A lever is a rigid rod that can rotate around a fixed point called fulcrum.

a) Give an example of a second order lever

b) Draw a diagram of a second order lever and label its parts.

c) Why is the mechanical advantage of second order levers always greater than one?

28. The weight of the object was found using a spring balance to find the buoyant force. The spring balance was damaged when it detected the weight of the object in the water. Explain how to find weight of object in water using meter scale, thread, weights and stand .

29. a) Which order of lever is scissors?

b) Is the mechanical advantage of a scissor always same ? Justify your answer?

30.



a) Which order of lever does the figure represent?

b) What is its mechanical advantage?

(less than one, one, between one and two, two)

c) Write another example for the same order lever?

31. A carpenter needs to split a log of wood using a wedge. The wedge has a length of 30 cm and a width of 6 cm.

a) Calculate the mechanical advantage of the wedge.

b) Suggest a modification to increase the mechanical advantage

32. A force of 5 N is applied to the effort arm of a second-order lever. The effort arm is 10 cm long, and the load arm is 4 cm long,

a) Calculate the mechanical advantage of the lever.

b) If a force of 80 N applied, calculate the load lifted by the lever.

c) Write an example of a second order lever.

33. A worker uses a fixed pulley to lift a load of 50 N.

a) What is the effort required to lift the load?

b) Explain how fixed pulley and movable pulley differ in their actions.

34. Match Appropriately

First order lever	- Mechanical advantage is less than one	- Hairpin roads
Inclined plane	- Mechanical advantage is one	- Fishing pole
Third order lever	- Mechanical advantage is greater than one	- seesaw

35. Inclined planes are often used in situations where objects are to be lifted.

a) What is the work done to lift a 1000 N load by 5 m?

b) To reduce the applied force to 100N , what is the length of the inclined plane needed?

c) Find the mechanical advantage of the inclined plane

4 Mark

36. a) Draw a schematic diagram of a first order lever with mechanical advantage more than one. Mark the fulcrum, load and effort?
    b) What will be the change in the mechanical advantage if the effort moves closer to the fulcrum?
    c) Write an example for a first order lever with mechanical advantage greater than one.

37. When effort is applied on a movable pulley, the pulley and the load will be lifted upwards along the other side of the rope.
    a) A movable pulley can be considered which type of lever?
    b) What is the mechanical advantage of a movable pulley?
    (Less than one, 1, Between one and two, Two)
    c) How much will the load rise if the rope is pulled 4 m by applying effort?
    d) What is the gain in work when using a movable pulley?

38. If a force of 3000 N is required to move an object to a height of 3 m through an inclined plane of length 9 m,
    a) What is the weight of the object?
    b) What is the mechanical advantage?
    c) Prove that the work done on the object and the work done by the effort are equal

39. Levers are useful for lifting heavy rocks easily.
    a) Which type of lever is suitable for lifting heavy objects and why?
    b) Calculate the mechanical advantage of a lever with an effort arm length of 2m and a load arm length of 0.5m.
    c) What is the force required to lift an object weighing 100 N?

40. A heavy statue is to be loaded onto a truck of 1.5 meters high. A 6-meter long wooden plank is used as a ramp to push the statue up.
    a) Calculate the mechanical advantage of the ramp

b) If the effort required to push the statue is 500 N, calculate the weight of the statue?

c) State one reason why using the ramp is better than lifting the statue directly.

d) Suggest a way to make this task easier?

41. Arun is helping his father lift a heavy bucket of water from a well.

a) Which order of lever is the pulley fixed at the top of the well.

b) If the bucket weighs 100 N, calculate the effort required using a single movable pulley.

c) What happened to the displacement if the effort reduced to half.

d) Explain why a movable pulley is more useful than a fixed pulley for lifting the heavy load.

42. A person is using a crowbar to lift a heavy rock stuck in the mud. It's a 2m long crowbar, with the fulcrum 20cm from the rock.

a) Which order of lever is crowbar?

b) Calculate the mechanical advantage of the crowbar.

c) If the person applies an effort of 50 N, calculate the maximum weight of the rock that can be lifted.

d) Explain why a longer crowbar would make it easier to lift the rock?

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Chapter 7

MECHANICAL ADVANTAGE IN ACTION

ANSWER KEY

1 Mark

1. d) All
2. a) The statement and the reason are correct, the reason explains the statement
3. d) i and iv only
4. c) One
5. c) ii,iv only
6. b) Can change the direction of the applied force
7. c) The statement is correct and the reason is wrong
8. a) i, ii, iv only
9. b) i, ii, iii only
10. a) The statement and the reason are correct, the reason explains the statement
11. Lemon squeeze, nutcracker
12. b) Statement 1 is correct 2 is incorrect
13. a) The heavier child should sit closer to the load
14. a) Statement 1 and 2 are correct

2 Mark

15. 5
16. To increase the speed of the vehicle the toothed wheel attached to the engine must be connected to the small toothed wheel that helps to turn the tyre of the vehicle. Large toothed wheel is connected to the toothed wheel attached to the engine while going uphill.

17 a) iii) large wheel

b) When connected to a large toothed wheel, the speed of the vehicle decreases, but its efficiency to rotate increases.

18. Yes. A larger wheel radius increases the mechanical advantage .

19. Fixed pulley – i and iii

    Movable pulley – ii and iv

20. Yes. The work done on the object and the work done by the effort are equal,hence no gain in work achieved while using a movable pulley.

21. 4m inclined plane. using a longer inclined plane provide more mechanical advantage.

22. a) First order lever

    b) Effort arm = 1.5m - 0.5m = 1.0m

    Load arm = 0.5m

    mechanical advantage = $\frac{1.0}{0.5} = 2$

23. a) Mechanical advantage (MA) is the ratio of the load to the effort. It is a number indicating how many times of the load is the effort.

    $$\text{Mechanical advantage, MA} = \frac{\text{Load}}{\text{Effort}}$$

    b) Examples: pulley, scissor

24. Mechanical advantage = $\frac{\ell}{h} = \frac{6}{1} = 6$

25. $\frac{L}{E} = \frac{R}{r}$

    $E = \frac{60 x 4.5}{18} = 15N$

3 marks

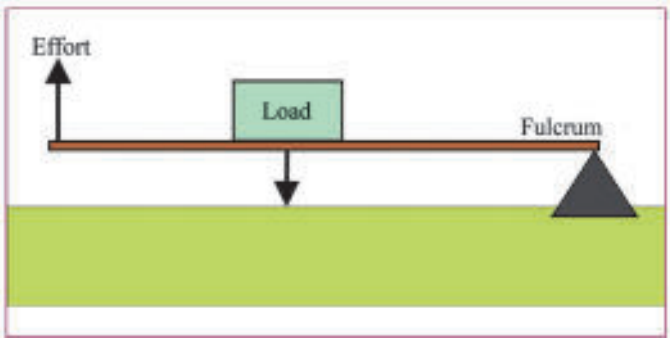
26. a) Second order lever

    b) greater than one

    c) Lemon squeezer.

27. a) Nutcracker

b)



c) Since effort arm > load arm, MA is always greater than 1.

28. Hang the meter scale on the centre of gravity. Hang the object submerged in water at a certain distance from that point. Hang a certain weight on the other side and balance the scale. Then the weight of the object can be determined using the equation load x load arm = effort x effort arm.

29. a) first order lever.

b) No. The length of the load arm varies depending on the use of the scissors. Accordingly, the mechanical advantage of the scissors varies.

30. a) First order lever

b) One

c) seesaw

31. a) Mechanical advantage = $\frac{\ell}{h} = \frac{30cm}{6cm} = 5$

b) Increase the length or reduce the width.

32. a) MA = 2.5.

b) 200N

c) Lemon squeezer

33. a) 50N

b) Difference: - Fixed pulley: Pulley rotates around a stationary axe. Only changes direction of force. MA = 1, effort = load.

- Movable pulley: Pulley moves with load; reduces effort needed. MA = 2,

34. First order lever - Mechanical advantage is one - seesaw

Inclined plane - Mechanical advantage is greater than one - Hairpin roads

Third order lever - Mechanical advantage is less than one - Fishing pole

35. a) a) Work done = Force x Displacement

1000 N x 5m = 5000J

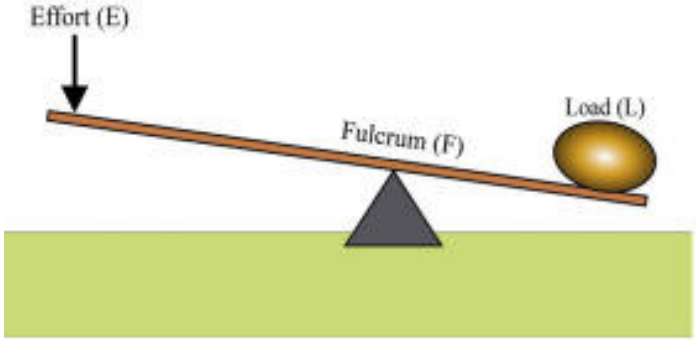
b) Work = Force x Displacement

Displacement = $\frac{5000}{100} = 50m$

c) Mechanical advantage = $\frac{\ell}{h} = \frac{50m}{5m} = 10$

4 marks

36. a)



b) Mechanical advantage will decrease

c) Crowbar

37. a) Secondorder lever

b) Two

c) Load rises 2m (half of rope pulled).

d) No gain in work

38. a) 9000N

b) 3

c) work done by the effort

= $E\ell = 3000x9 = 27000\,\text{J}$

work done by the load

= Lh = 9000x3 = 27000J

39. a) Second order lever and first order lever with greater lever arm.

effort arm > load arm ; MA> 1.

b) MA = $\dfrac{\text{Effort arm}}{\text{Load arm}} = \dfrac{2m}{0.5m} = 4$

c) Effort = $\dfrac{\text{Load}}{M.A} = \dfrac{100N}{4} = 25N$

40. a) $M.A = \dfrac{\ell}{h} = \dfrac{6}{1.5} = 4$

b) Load = Effort x M.A = 500N x 4 = 2000N

c) Requires less force,more mechanical advantage.

d) Use a longer inclined plane

.41. a) Fixed pulley (single wheel fixed at top).

b) Effort = 50N

c) Displacement doubles (load moves half the effort distance).

d) Movable pulley is more useful because it halves the effort needed (MA = 2), making lifting easier.

42. a) First order lever.

b) MA = 9

c) 450N

d) Longer crowbar increases effort arm, raising MA, making lift easier.

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